

The Most Informative Bit and Beyond: A Proof of the Courtade–Kumar Conjecture and Multibit Extensions

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Abstract—Let $X = (X_1, \dots, X_n)$ be uniform on the Boolean cube, and let Y be obtained by passing its coordinates independently through a binary symmetric channel with crossover probability p . A natural quantization question asks how much information about Y can be retained by a function $Q(X)$ constrained to k output bits. For $k \leq n$, reporting k input coordinates retains $k(1 - h_2(p))$ bits of information, where h_2 denotes the binary entropy function. This makes coordinate projections the natural benchmark.

The case $k = 1$ is the Courtade–Kumar conjecture, which posits that every Boolean function f satisfies

$$I(f(X); Y) \leq 1 - h_2(p),$$

with equality attained by coordinate functions. We prove this conjecture for every dimension and crossover probability, without any balance assumption on f . Our proof views the gap relative to the coordinate benchmark as a quantity that evolves with the channel correlation. After a monotone rearrangement, an entropy-flow identity relates its derivative to the edge boundary of the corresponding Boolean decision set. Combining a sharp local entropy comparison with Fourier analysis on the Boolean cube and logarithmic Sobolev estimates for pivotal sets yields a global differential inequality: if the gap is positive at any correlation level, then it must grow as the noise decreases. This contradicts the noiseless endpoint, where the gap is nonpositive.

For $k \geq 8$, however, a distinctly coding-theoretic behavior emerges. We construct quantizers from shortened Hamming codes whose covering structure beats the coordinate-projection benchmark for $k = 8$, and a direct-sum argument extends this failure to every $k \geq 8$. Thus, unlike in the single-bit case, structured coding can retain more information than simple coordinate selection. Finally, searches within structured code families for $2 \leq k \leq 7$, including exhaustive checks in selected families, and Monte Carlo searches over seven-bit random codebooks find no violation below eight bits. These findings lead us to conjecture that coordinate projections remain optimal for $2 \leq k \leq 7$.

Index Terms—Binary symmetric channel, Boolean functions, Courtade–Kumar conjecture, Fourier analysis, Hamming codes, logarithmic Sobolev inequality, mutual information, quantization.

I. INTRODUCTION

Let $X = (X_1, \dots, X_n)$ be a sequence of independent uniform binary random variables, and let Y be obtained by passing every coordinate through a binary symmetric channel (BSC) with crossover probability $p \in [0, 1/2]$. A Boolean

summary $f(X)$ retains one bit from the input. Courtade and Kumar conjectured that the mutual information between this summary and the complete noisy observation is maximized by retaining one of the original coordinates [1]. Equivalently,

$$I(f(X); Y) \leq 1 - h_2(p), \quad (1)$$

where h_2 is the binary entropy function. Equality is attained by any *dictator* defined as $f(X) = X_i$ for some $i \in \{1, 2, \dots, n\}$.

The problem is different from maximizing correlation or noise stability. The observation Y is an n -dimensional random vector, and information carried jointly by its coordinates need not decompose into coordinatewise contributions. For example, the parity of two noisy bits can contain information that is absent from either noisy bit alone. Consequently, results for two Boolean outputs [2], [3] or for sums of coordinatewise mutual informations [4] do not directly imply (1). The output $f(X)$ may also be biased, which removes several symmetries available in the balanced case.

This paper establishes (1) in full generality. The proof uses the correlation parameter $\rho = 1 - 2p$. After monotone sorting, the mutual information becomes an entropy functional of the posterior mean $T_\rho f$. Its derivative admits an edge decomposition. The central step is to strengthen the entropy-constrained edge envelope by retaining the squared angular displacement lost in a Cauchy–Schwarz comparison. The pivotal sets of the same Boolean function supply the Fourier and logarithmic-Sobolev control needed to pay for this correction. The final argument is dynamical: wherever a function has more information than a dictator, its advantage must increase with ρ . Such an advantage cannot return to its nonpositive value at $\rho = 1$.

A Lean 4 formalization of our one-bit proof, together with its definitions, pinned dependencies, and verification instructions, is available in the project’s `lean` directory.¹ Its exported statement uses finite-distribution mutual information in bits and covers every crossover probability $p \in [0, 1]$.

We also study the natural vector-valued extension in which $f(X)$ contains k bits. The scalar theorem does not tensorize through the chain rule because conditioning on preceding output bits destroys the uniform input law. In fact, the coordinate benchmark is false for sufficiently large fixed k . We give explicit shortened-Hamming counterexamples for $k = 8, 9, 10$, extend them to every $k \geq 8$, and formulate a threshold conjecture for $2 \leq k \leq 7$.

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¹<https://github.com/hessammahdaviyar/most-informative-bit/tree/main/lean>

A. Related Work

Courtade and Kumar introduced the full-vector problem and proved several structural reductions, including the monotone sorting operation used below [1]. The conjecture stimulated a sequence of Fourier, functional-inequality, and information-geometric approaches. The Friedgut–Kalai–Naor theorem identifies the rigidity of Boolean functions whose Fourier mass is concentrated on the first two levels [5]. Samorodnitsky established the conjecture in a dimension-free high-noise regime and developed strengthened inequalities for the entropy of noisy Boolean functions [6], [7]. Ordentlich, Shayevitz, and Weinstein obtained sharper balanced high-noise bounds using Fourier analysis and hypercontractivity [8], while Yang and Wesel gave a complementary calculus argument in a dimension-dependent neighborhood of complete noise [9].

Yu developed a general Φ -stability framework and subsequently proved broad local-optimality and balanced finite-range results using quantitative Fourier estimates [10]–[12]. Javanmard and Woodruff proved the dictator bound for the sum of the individual-coordinate mutual informations and sharpened the high-noise analysis [4]; the related Boolean-hull and moment relaxations are discussed in [13]. Li and Médard connected the problem to posterior moments and noninteractive correlation distillation [14], and Barnes and Özgür showed that the balanced conjecture is essentially equivalent to a symmetrized Li–Médard formulation [15].

A second line of work seeks stronger entropy or isoperimetric statements. Anantharam, Bogdanov, Chakrabarti, Jayram, and Nair proposed a Hellinger inequality implying the conjecture [16]. Chen and Nair placed this proposal in an ordered family of entropy inequalities and related the low-noise limits to sharp cube isoperimetry [17]. The critical square-root isoperimetric inequality was recently established by Durcik, Ivanisvili, Roos, and Xie [18], confirming the balanced low-noise limit predicted by the Hellinger conjecture. This limiting result does not by itself establish the finite-noise Shannon inequality. Other approaches use principal inertia components [19], adjacency events [20], Gaussian analogues [21], and stochastic interpolation [22].

The closest antecedent to the present proof is the differential-equation framework of Chen, Gohari, and Nair [23], introduced in 2025. Their entropy-flow identity and entropy-constrained two-point extremal formula are used here.

Three concurrent works also establish the scalar conjecture in full generality [24]–[26]. Chen, Gohari, Javanmard, Lin, Mirrokni, Nair, and Woodruff close the differential-equation program through an unrestricted four-moment Bellman inequality for a hybrid potential, using analytic reductions together with certified interval arithmetic. Ky and Tran instead develop an entropy-production and spectral framework. Their method is substantially closer to ours: both arguments use monotone compression, edgewise entropy-production estimates, Fourier control of functions with diffuse coordinate dependence, separate channel regimes, and a differential contradiction at the noiseless endpoint. The quantitative implementations differ. Our proof centers on an angular edge correction and logarithmic-Sobolev estimates for pivotal sets, whereas Ky and Tran use

a profile-clock integration and selected-coordinate spectral estimates. In another concurrent work, Kramer and Saglam [26] give a substantially shorter proof based on induction on the dimension, applied to a strengthened inequality that retains the output mean. We have independently verified their proof in Lean. Finally, neither of [24]–[26] considers the k -bit extension studied in the present paper.

As mentioned above, the present paper develops the vector-valued extension in which the quantizer reports k bits. For this problem, Chandar and Tchamkerten characterized the positive-rate asymptotics and gave Hamming- and Golay-code counterexamples with $k \geq 11$ [27]. Huleihel and Ordentlich proved coordinate optimality at the complementary endpoint $k = n - 1$ [28]. The finite-block behavior observed in our code search is also related qualitatively to recent results showing that structured Reed–Solomon-based codebooks can outperform random codebooks under average Hamming covering radius [29]; the objective in that work is different from BSC mutual information.

B. Contributions and Organization

Our first main contribution is Theorem 7, which proves the Courtade–Kumar conjecture in every dimension and for every crossover probability, without any balance assumption on the Boolean function. The proof follows a single dynamical strategy. In natural units, set $I_f(\rho) := I(f(X); Y_\rho)$, $\phi(\rho) := I(X_1; Y_\rho)$, $\Delta_f(\rho) := I_f(\rho) - \phi(\rho)$. We show that a positive advantage over a coordinate projection would have to grow strictly as the channel becomes less noisy. Section II introduces the posterior representation of I_f , the required Fourier notation, and the monotone reduction. In particular, Lemma 1 shows that coordinatewise sorting preserves any advantage, allowing the remainder of the proof to exploit the edge structure of a monotone Boolean decision set. An entropy-flow identity expresses $\rho I'_f(\rho)$ as an action functional of the posterior mean $T_\rho f$. Section III then decomposes this action over edges of the discrete cube. The key estimate, Lemma 2, strengthens the usual two-point entropy envelope by retaining a squared angular correction that is exact for dictators. The rest of the proof shows that the Boolean and spectral structure of f always provides enough reserve to pay for this correction.

Writing $\ell = \operatorname{arctanh} \rho$, we divide the channel into the small-correlation range $0 \leq \ell \leq 31/20$, the intermediate range $31/20 \leq \ell \leq 7/2$, and the near-noiseless tail $\ell \geq 7/2$. Equivalently, the two junctions are $\rho = \tanh(31/20)$ and $\rho = \tanh(7/2)$. These cutoffs are technical junctions at which the corresponding estimates meet, rather than asserted phase transitions in the problem. Proposition 8 settles the small-correlation range directly, at every output bias, by combining a Fourier energy gap with scalar entropy bounds. In the intermediate range, the pivotal sets of f supply the needed reserve: logarithmic Sobolev inequalities control the noisy entropies of their indicators, and a log-sum argument combines the coordinatewise costs with the global edge budget. This yields Proposition 3, according to which $\Delta_f(\rho) > 0$ implies $\Delta'_f(\rho) > 0$ throughout that range. The bounded angular reserve is insufficient by itself near the noiseless

endpoint, so the tail argument supplements the same edgewise inequality with degree-weighted Fourier information. Lemma 4 identifies the relevant coordinate deficits, Lemma 5 converts the information advantage into an entropy-funded spectral baseline, and Lemma 6 turns the resulting budget comparison into strict information growth. Proposition 9 separately rules out a counterexample having a sufficiently influential coordinate. Section V assembles these complementary estimates in Lemma 10, which gives the global implication

$$\Delta_f(\rho) > 0 \implies \Delta'_f(\rho) > 0.$$

Since $\Delta_f(1) = H(f(X)) - \log 2 \leq 0$, a positive advantage cannot persist to the noiseless endpoint; this contradiction completes the proof of Theorem 7.

Our second contribution, developed in Section VI, concerns the natural extension in which the quantizer reports k bits and the coordinate benchmark is $k(1 - h_2(p))$. Proposition 11 shows that affine linear maps cannot violate this benchmark. Nonlinear nearest-codeword quantizers built from linear codebooks behave differently: an exact high-noise expansion, expressed through the coset-leader distribution, produces shortened-Hamming counterexamples for $k = 8, 9, 10$; Proposition 12 extends the failure to every $k \geq 8$ by a direct-sum construction. Below eight bits, structured-code searches and seven-bit random-codebook experiments find no violation. The seven-bit examples also show an advantage of the tested structured codebooks over the sampled random codebooks. These findings motivate Conjecture 13, which predicts that coordinate projections remain optimal for $2 \leq k \leq 7$. The appendices establish the scalar and Fourier bounds, endpoint reductions, and channel comparisons needed for the proof.

II. PRELIMINARIES

We fix the entropy and Fourier normalizations used in the one-bit proof. The posterior representation turns mutual information into a function of the channel correlation, and monotone reduction permits the edge and pivotal-set estimates that follow.

A. Notation

Let X be uniform on $\{-1, 1\}^n$, and let Y be its observation through a BSC with crossover probability $(1 - \rho)/2$, where $\rho \in [0, 1]$. Expectations are uniform on the appropriate cube unless otherwise specified. Throughout the one-bit proof and its appendices, logarithms are natural and information is measured in nats. For $r \in [-1, 1]$, define

$$\phi(r) = \frac{1+r}{2} \log(1+r) + \frac{1-r}{2} \log(1-r),$$

$$L = \log 2, \quad h(r) = L - \phi(r).$$

Thus $h(r)$ is binary entropy in the mean parametrization. We use the convention $0 \log 0 = 0$. Write $g(r) = \phi'(r) = \operatorname{atanh} r$ for $|r| < 1$, and $\ell_\rho = \operatorname{atanh} \rho$ for $0 < \rho < 1$.

For $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, let

$$m = \mathbb{E}f, \quad u(y) = T_\rho f(y) = \mathbb{E}[f(X) \mid Y = y].$$

Then

$$I_f(\rho) := I(f(X); Y) \\ = h(m) - \mathbb{E}h(u) = \mathbb{E}\phi(u) - \phi(m). \quad (2)$$

A signed dictator $f(x) = \pm x_i$ attains $I_f(\rho) = \phi(\rho)$. Write $\Delta_f(\rho) = I_f(\rho) - \phi(\rho)$ for the information gap. Constant functions have zero mutual information; in the one-bit argument, assume f is nonconstant, so $|u(y)| < 1$ for $0 < \rho < 1$.

B. Fourier Analysis and Entropy Flow

We use the standard Boolean Fourier conventions [30]. For $S \subseteq [n]$, let $\chi_S(x) = \prod_{i \in S} x_i$ and $\hat{v}(S) = \mathbb{E}[v\chi_S]$. For any real-valued cube function v ,

$$v = \sum_{S \subseteq [n]} \hat{v}(S) \chi_S, \quad \mathbb{E}v^2 = \sum_S \hat{v}(S)^2,$$

$$T_\rho v = \sum_S \rho^{|S|} \hat{v}(S) \chi_S.$$

The cube Laplacian is normalized by

$$\mathcal{L}\chi_S = |S| \chi_S,$$

$$\mathcal{L}v(x) = \frac{1}{2} \sum_{i=1}^n (v(x) - v(x^{\oplus i})),$$

where $x^{\oplus i}$ denotes x with coordinate i flipped. For fixed f and $0 < \rho < 1$, $\rho \partial_\rho u = \mathcal{L}u$, and the entropy-flow identity [23, Lemma 1] gives

$$A(u) := \mathbb{E}[g(u)\mathcal{L}u] = \rho I'_f(\rho). \quad (3)$$

C. Monotone Reduction

We use the following consequence of coordinate compression [1, Lemma 2 and Remark 2].

Lemma 1 (Monotone reduction). *For every Boolean f , there is a coordinatewise nondecreasing Boolean $\tilde{f}$ in the same dimension such that $\mathbb{E}\tilde{f} = \mathbb{E}f$ and $I_{\tilde{f}}(\rho) \geq I_f(\rho)$ for all $\rho \in [0, 1]$.*

It therefore suffices to consider a fixed nonconstant monotone function throughout the one-bit proof.

III. EDGE GEOMETRY AND THE PIVOTAL DECOMPOSITION

Fix a nonconstant monotone f , let $0 < \rho < 1$, and write $u = T_\rho f$. We bound the information growth $A(u)$ using the geometry of the pivotal sets. The key estimate combines the Dirichlet energy of $\Theta = \arcsin u$ with an entropy-dependent correction. The energy term will be controlled by Fourier analysis.

Let $f_{i,\pm}$ denote the sections of f with coordinate i fixed to ± 1 . Define the pivotal indicator and its mean by

$$q_i^f = \frac{f_{i,+} - f_{i,-}}{2} \in \{0, 1\},$$

$$a_i = \mathbb{E}q_i^f = \hat{f}(\{i\}) = \operatorname{Inf}_i(f).$$

The indicators q_i^f will also enter the logarithmic Sobolev estimates below. All coordinate sums in this section omit indices with $a_i = 0$.

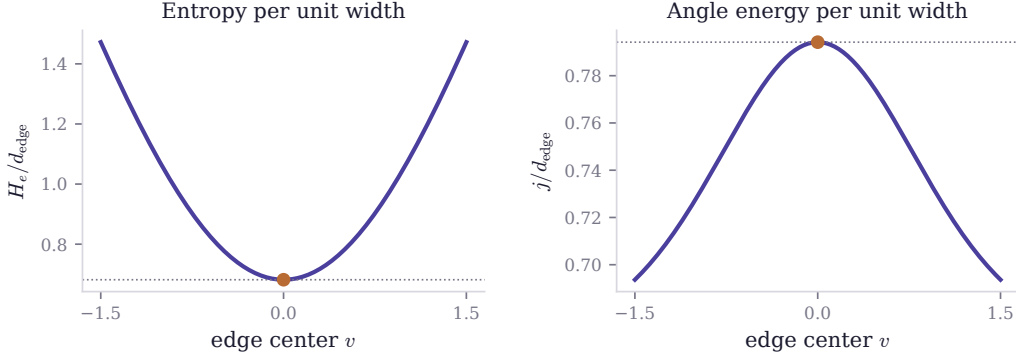


Fig. 1. Centered-edge comparison for $p = \tanh(v + k)$, $q = \tanh(v - k)$, and $k = 0.8$. Here $d_{\text{edge}} = |p - q|/2$, $H_e = [h(p) + h(q)]/2$, and $j = (\arcsin p - \arcsin q)^2/4$. Centering at $v = 0$ minimizes H_e/d_{edge} and maximizes j/d_{edge} , yielding the scalar bounds used in Lemma 2.

For $v > 0$, define

$$F(v) = \frac{h(\tanh v)}{\tanh v}, \quad c(v) = \frac{(\arcsin(\tanh v))^2}{\tanh v}.$$

The function F is strictly decreasing from infinity to zero. A common entropy budget determines the comparison heights in the following two bounds. Figure 1 illustrates the underlying centered-edge comparison.

Lemma 2 (The edge budget). *Let $H_{\text{budget}} \geq \mathbb{E}h(u) > 0$. For each i with $a_i > 0$, let $v_i > 0$ be the unique solution of*

$$F(v_i) = \frac{H_{\text{budget}}}{\rho a_i}.$$

Then

$$A(u) \geq \rho \sum_i a_i v_i, \quad (4)$$

$$A(u) \geq \mathbb{E}[\Theta \mathcal{L} \Theta] + \rho \sum_i a_i (v_i - c(v_i)). \quad (5)$$

The first inequality is the entropy-constrained edge bound [23, Theorem 4 and Appendix C]. Both bounds follow by averaging convex scalar edge inequalities; the full proof is given in Appendix A.

Whenever $\mathbb{E}h(u) \leq h(\rho)$, the choice $H_{\text{budget}} = h(\rho)$ defines heights ℓ_i satisfying

$$F(\ell_i) = \frac{F(\ell_\rho)}{a_i}, \quad 0 < \ell_i \leq \ell_\rho. \quad (6)$$

The first bound gives

$$W := \sum_i a_i \ell_i \leq \frac{A(u)}{\rho}. \quad (7)$$

The second bound will be combined with a spectral estimate for $\mathbb{E}[\Theta \mathcal{L} \Theta]$, retaining the correction term. The general budget formulation also permits the sharper entropy bound used in the intermediate-correlation argument.

IV. INFORMATION GROWTH ACROSS CHANNEL REGIMES

This section supplies the differential step of the proof: a positive information gap forces strictly positive growth. In the intermediate range, pivotal entropy controls the angular energy

in the edge budget. Near the noiseless endpoint, coordinatewise spectral deficit bounds replace that estimate; both arguments retain the same angular comparison. The small-correlation bound and the final contradiction are assembled in Section V.

A. Intermediate-Correlation Growth

Write $r = \rho$, $\ell = \text{atanh } r$, and $\Delta_f(r) = I_f(r) - \phi(r)$. We combine an angular-energy bound with logarithmic Sobolev control of pivotal sets, then close the edge budget using the log-sum inequality.

Proposition 3 (Intermediate-correlation growth). *For every monotone Boolean f and $31/20 \leq \ell \leq 7/2$,*

$$\Delta_f(r) > 0 \implies I'_f(r) > \phi'(r).$$

1) *Angular-energy bound:* Set $J_{\text{ang}} = \mathbb{E}[\Theta \mathcal{L} \Theta]$, where $\Theta = \arcsin u$, and define

$$\begin{aligned} \theta &= \arcsin r, & \lambda &= \theta/r, \\ \alpha &= \frac{\theta + \sinh \ell}{\ell}, & \gamma &= \lambda \alpha, \\ \eta &= \frac{\theta^2 \gamma}{\gamma + 1}. \end{aligned}$$

For $\ell \geq 31/20$ and $x \in [-1, 1]$, Appendix B gives

$$\frac{\alpha}{2} (\arcsin x - \lambda x)^2 \leq x \arcsin x - r \theta - \alpha [\phi(x) - \phi(r)]. \quad (8)$$

Averaging at $x = u$, multiplying by 2λ , and completing the square in $\mathcal{L} + \gamma$ yields

$$J_{\text{ang}} \geq 2\theta^2 + 2\gamma [\phi(m) + \Delta_f(r)] + \lambda^2 \mathbb{E} [u \{ \gamma - (\gamma + 1)^2 (\mathcal{L} + \gamma)^{-1} \} u]. \quad (9)$$

2) *Pivotal entropy:* For $a_i > 0$ and $0 \leq t \leq r^2$, let

$$w_i(t) = T_{\sqrt{t}} q_i^f, \quad s_i(t) = \mathbb{E} w_i(t)^2,$$

with noise acting on the other coordinates. All sums below run over $a_i > 0$. Define

$$z_i = \frac{\gamma + 1}{\gamma r^2} \int_0^{r^2} [1 - (t/r^2)^\gamma] s_i(t) dt, \quad Z = \sum_i z_i,$$

and put $V_0 = 1 - m^2$, $U = \text{Var}(u)/r^2$. The integral is a probability average, so $a_i^2 \leq z_i \leq a_i$. Fourier expansion gives

$$Z = (\gamma + 1) \sum_{k \geq 1} \frac{r^{2(k-1)}}{\gamma + k} \sum_{|S|=k} \hat{f}(S)^2, \quad 0 < Z \leq U \leq V_0.$$

Gross's logarithmic Sobolev inequality [30], [31], together with Jensen's inequality, gives

$$\mathbb{E}[w_i \mathcal{L} w_i] \geq \frac{1}{2} s_i \log \frac{s_i}{a_i^2}.$$

Averaging with the kernel defining z_i , applying convexity of $s \log(s/a_i^2)$, and expanding in Fourier degrees yields

$$(\gamma + 1)(U - Z) \geq \frac{1}{2} E_{\text{piv}}, \quad E_{\text{piv}} := \sum_i z_i \log \frac{z_i}{a_i^2}. \quad (10)$$

With

$$M(m) = 2\gamma\phi(m) - [\lambda^2(2 - r^2) + \lambda/\alpha]m^2,$$

combining (10) with the Fourier expansion of (9) gives

$$J_{\text{ang}} \geq \theta^2(1 + V_0 - Z) + \frac{\eta}{2} E_{\text{piv}} + M(m) + 2\gamma\Delta_f(r). \quad (11)$$

3) *Closing the edge budget:* Assume $31/20 \leq \ell \leq 7/2$, $\Delta = \Delta_f(r) > 0$, and $A(u) \leq r\ell$. Monotonicity gives $a_i \leq 1 - |m| \leq V_0$. Since $\phi(m) \geq m^2/2$ and $h(r) < 1/2$ in this range,

$$\mathbb{E}h(u) = h(r) - \phi(m) - \Delta \leq V_0 h(r).$$

Apply Lemma 2 with $H_{\text{budget}} = V_0 h(r)$. Its heights satisfy

$$F(v_i) = \frac{V_0 F(\ell)}{a_i}, \quad 0 < v_i \leq \ell.$$

Writing $H(v) = v - c(v) \geq 0$, we obtain

$$\sum_i a_i v_i \leq \ell, \quad A(u) \geq J_{\text{ang}} + r \sum_i a_i H(v_i). \quad (12)$$

Put $\beta = 2r/\eta$. Lemma 16 and $2\phi(m) \geq m^2$ imply

$$\log \frac{vF(v)}{\ell F(\ell)} \geq \beta[H(\ell) - H(v)], \quad 0 < v \leq \ell, \quad (13)$$

$$\theta^2 - rH(\ell) - \eta/2 \geq 0, \quad (14)$$

$$M(m) - rH(\ell)m^2 \geq 0. \quad (15)$$

Set $b_i = a_i^2 e^{-\beta H(v_i)}$ and $B_{\text{piv}} = \sum_i b_i$. By (13) and (12),

$$B_{\text{piv}} \leq \frac{V_0}{\ell} e^{-\beta H(\ell)} \sum_i a_i v_i \leq V_0 e^{-\beta H(\ell)}.$$

Using $z_i \leq a_i$ and $\eta\beta/2 = r$, the log-sum inequality therefore gives

$$\begin{aligned} \frac{\eta}{2} E_{\text{piv}} + r \sum_i a_i H(v_i) &\geq \frac{\eta}{2} \sum_i z_i \log \frac{z_i}{b_i} \\ &\geq \frac{\eta}{2} Z \log \frac{Z}{B_{\text{piv}}} \\ &\geq \frac{\eta}{2} Z \log \frac{Z}{V_0} + rZH(\ell). \end{aligned}$$

Combining this with (11) and (12), then using $r\ell = \theta^2 + rH(\ell)$ and $Z \log(Z/V_0) \geq Z - V_0$, yields

$$A(u) - r\ell \geq [\theta^2 - rH(\ell) - \eta/2](V_0 - Z) + M(m) - rH(\ell)m^2 + 2\gamma\Delta > 0,$$

where the strict inequality follows from (14)–(15) and $\Delta > 0$. This contradicts $A(u) \leq r\ell$ and proves the proposition by (3).

B. High-Correlation Spectral Bounds

For $\ell = \text{atanh } \rho \geq 7/2$, we estimate the spectral term in (9) using a reference Fourier degree. Retain $\gamma = \lambda\alpha$ and let $d \geq 2$ be real; the tail argument uses $d = \ell/2 + 5/4$. Define

$$\begin{aligned} \omega(s) &= \frac{\gamma s}{\gamma + s}, \\ K_{\text{LS}} &= \frac{\gamma^2 e^{2d-3}}{2(\gamma + d)(\gamma + 1)}, \\ d_j &= [\omega(d) - \omega(j)]\rho^{2j}, \quad j = 1, 2. \end{aligned}$$

Lemma 4 (Coordinate deficit bounds). *Let $u = T_\rho f$, where f is monotone Boolean and $0 < \rho < 1$. Let $\gamma > 0$, $d \geq 2$, and set*

$$D_i = \sum_{S \ni i} \frac{\omega(d) - \omega(|S|)}{|S|} \hat{u}(S)^2,$$

$$D(a) = \min \left\{ K_{\text{LS}} \rho^2 a^2, \frac{d_2}{2} a + \left(d_1 - \frac{d_2}{2} \right) a^2 \right\}.$$

Then $D_i \leq D(a_i)$, and

$$\sum_{S \neq \emptyset} \omega(|S|) \hat{u}(S)^2 \geq \omega(d) \text{Var}(u) - \sum_i D(a_i). \quad (16)$$

Proof. The case $a_i = 0$ is immediate. For $w \geq 0$, write $b = \mathbb{E}w > 0$ and $s = \mathbb{E}w^2$. The cube logarithmic Sobolev inequality [30], [31] and Jensen's inequality give

$$\begin{aligned} \mathbb{E}[w(\mathcal{L} + 1 - d)w] &\geq \frac{s}{2} \log \frac{s}{b^2} + (1 - d)s \\ &\geq -\frac{1}{2} e^{2d-3} b^2, \end{aligned}$$

where the last step minimizes over $s > 0$.

Set $p_i = (u_{i,+} - u_{i,-})/2$ and $w_t = T_{\sqrt{t}} p_i$, with noise and $\mathcal{L}_{-i}$ acting on the remaining coordinates. Then $w_t \geq 0$ and $\mathbb{E}w_t = \rho a_i$. Fourier expansion gives

$$D_i = \frac{\gamma}{\gamma + d} \int_0^1 (1 - t^\gamma) \mathbb{E}[w_t(d - 1 - \mathcal{L}_{-i})w_t] dt,$$

using $\int_0^1 (1 - t^\gamma) t^{s-1} dt = \gamma/[s(\gamma + s)]$. The preceding estimate and $\int_0^1 (1 - t^\gamma) dt = \gamma/(\gamma + 1)$ yield $D_i \leq K_{\text{LS}} \rho^2 a_i^2$.

For the second bound, Parseval applied to the pivotal indicator q_i^f gives $\sum_{S \ni i} \hat{f}(S)^2 = a_i$, with singleton contribution a_i^2 . For $s \geq 2$, $[\omega(d) - \omega(s)]\rho^{2s} \leq d_2$: both nonnegative factors decrease on $2 \leq s \leq d$, and the product is nonpositive for $s > d$. Hence

$$D_i \leq d_1 a_i^2 + \frac{d_2}{2} (a_i - a_i^2),$$

which proves the second branch of $D(a_i)$. Finally, $\sum_i D_i = \sum_{S \neq \emptyset} [\omega(d) - \omega(|S|)] \hat{u}(S)^2$; summing the coordinate bounds proves (16). $\square$

C. Spectral Baseline

We now combine the angular comparison with the coordinate deficit bounds. The information gap supplies a lower bound on the noisy second moment, leaving only scalar channel conditions to control the output bias.

Set

$$\begin{aligned} p &= (\lambda + 1/\alpha)^2, & b &= \lambda^2(2 + 1/\gamma), \\ C_d &= p\omega(d) - b, \\ B &= 2\theta^2 + C_d \frac{\phi(\rho)}{L}, & k_\Delta &= 2\gamma + \frac{C_d}{L}. \end{aligned}$$

Lemma 5 (Spectral baseline). *Let $u = T_\rho f$ for monotone Boolean f , with $0 < \rho < 1$, $\ell = \operatorname{atanh} \rho \geq 31/20$, $\gamma = \lambda\alpha$, and $d \geq 2$. If*

$$C_d \geq 0, \quad \gamma - b + C_d \left(\frac{1}{2L} - 1 \right) \geq 0, \quad (17)$$

then

$$\mathbb{E}[\Theta \mathcal{L} \Theta] \geq B - p \sum_i D(a_i) + k_\Delta \Delta_f(\rho). \quad (18)$$

Proof. Since $\gamma = \lambda\alpha$,

$$\lambda^2 \left[\gamma - \frac{(\gamma + 1)^2}{\gamma + s} \right] = p\omega(s) - b.$$

Substitute this into (9) and apply Lemma 4. Using $C_d \geq 0$ and

$$\mathbb{E}\phi(u) = \phi(\rho) + \phi(m) + \Delta_f(\rho) \leq L\mathbb{E}u^2$$

gives

$$\begin{aligned} \mathbb{E}[\Theta \mathcal{L} \Theta] &\geq B - p \sum_i D(a_i) + k_\Delta \Delta_f(\rho) \\ &\quad + k_\Delta \phi(m) - (C_d + b)m^2. \end{aligned}$$

The last two terms have nonnegative sum by $\phi(m) \geq m^2/2$ and (17), proving the claim. $\square$

D. Yield Comparison and Growth

The two edge bounds now close the argument. We express each coordinate's angular cost and spectral loss per unit of edge weight, then compare this quantity with the common baseline.

For $0 < v \leq \ell = \operatorname{atanh} \rho$, define

$$\text{yield}(v) = \frac{\rho c(v)}{v} + \frac{pD(a)}{av}, \quad a = \frac{F(\ell)}{F(v)}.$$

Lemma 6 (Information growth). *Under the hypotheses of Lemma 5, suppose $\Delta_f(\rho) \geq 0$ and*

$$\frac{B}{\ell} \geq \rho, \quad \frac{B}{\ell} \geq \text{yield}(\ell_i) \quad \text{for all } i \text{ with } a_i > 0. \quad (19)$$

Then

$$\Delta'_f(\rho) \geq \frac{k_\Delta \ell}{B} \Delta_f(\rho). \quad (20)$$

In particular, $\Delta_f(\rho) > 0$ implies $\Delta'_f(\rho) > 0$.

Proof. Since $\Delta_f(\rho) \geq 0$, $\mathbb{E}h(u) = h(\rho) - \phi(m) - \Delta_f(\rho) \leq h(\rho)$. Lemmas 2 and 5, together with (19), therefore give

$$\begin{aligned} A(u) &\geq B + k_\Delta \Delta_f(\rho) + \sum_i a_i \ell_i [\rho - \text{yield}(\ell_i)] \\ &\geq B + k_\Delta \Delta_f(\rho) + \left(\rho - \frac{B}{\ell} \right) W. \end{aligned}$$

Using $W \leq A(u)/\rho$ and $\rho - B/\ell \leq 0$, we obtain

$$\frac{B}{\rho \ell} A(u) \geq B + k_\Delta \Delta_f(\rho).$$

Now $B/\ell \geq \rho > 0$ and $\Delta'_f(\rho) = A(u)/\rho - \ell \text{yield}$ (20). $\square$

V. PROOF OF THE MAIN THEOREM

We combine the direct small-correlation and influential-coordinate bounds with the growth estimates of Section IV. Together they show that a positive gap would have positive derivative at its interior maximum, completing the contradiction.

Theorem 7 (Main Theorem). *For every $n \geq 1$, every Boolean $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, and every $0 \leq \rho \leq 1$,*

$$I(f(X); Y) \leq \phi(\rho). \quad (21)$$

Dividing by $\log 2$ and setting $\rho = 1 - 2p$ gives (1) for $0 \leq p \leq 1/2$. Complementing every bit of Y preserves mutual information and replaces p by $1 - p$, so the same bound holds for $1/2 < p \leq 1$.

A. Direct Bounds

Set $\rho_* = \tanh(31/20)$.

Proposition 8 (Small correlation). *For every Boolean f and $0 \leq \rho \leq \rho_*$,*

$$I_f(\rho) \leq \phi(\rho).$$

Proof. See Appendix D. $\square$

Proposition 9 (Influential coordinate). *Let f be Boolean, $0 < \rho < 1$, and $a = |\hat{f}(\{i\})|$ for some coordinate i . If*

$$(1 - \rho^2)h(\rho a) \geq (1 - \rho^2 a)h(\rho), \quad (22)$$

then $I_f(\rho) \leq \phi(\rho)$.

Proof. Appendix A derives a mean-dependent form of Yu's local comparison [11, Theorem 4.6 and Remark 4.7]. $\square$

B. Global Growth Comparison

Lemma 10 (Growth of a positive gap). *For every monotone Boolean f and $0 < \rho < 1$,*

$$I_f(\rho) > \phi(\rho) \implies I'_f(\rho) > \phi'(\rho).$$

Proof. Assume $I_f(\rho) > \phi(\rho)$ and write $\ell = \operatorname{atanh} \rho$. Proposition 8 excludes $\ell \leq 31/20$, and Proposition 3 covers $31/20 \leq \ell \leq 7/2$.

For $\ell \geq 7/2$, Proposition 9 implies that (22) fails for every coordinate. Appendix E reduces the height comparison to the

half-height join and cutoff. With $\gamma = \lambda\alpha$ and $d = \ell/2 + 5/4$, Appendix F establishes (17) and

$$\frac{B}{\ell} \geq \max\{\rho, \text{yield}(\ell_i)\} \quad \text{for every } i \text{ with } a_i > 0.$$

Lemma 6 therefore gives $\Delta'_f(\rho) > 0$. $\square$

Proof of Theorem 7. Constants are immediate. By Lemma 1, it suffices to consider nonconstant monotone f . The function $\Delta_f = I_f - \phi$ is continuous on $[0, 1]$, differentiable on $(0, 1)$, and satisfies

$$\Delta_f(0) = 0, \quad \Delta_f(1) = -\phi(m) \leq 0.$$

If it were positive somewhere, it would attain a positive maximum at an interior point where $\Delta'_f = 0$, contradicting Lemma 10. $\square$

VI. VECTOR-VALUED QUANTIZATION: COUNTEREXAMPLES FROM EIGHT BITS

Having established the one-bit bound in Theorem 7, we ask whether a k -bit summary is optimized by retaining k input coordinates. Failure for large k is known [27]; here we give explicit counterexamples beginning at eight bits and examine the smaller-output regime.

Shortened Hamming quantizers violate the coordinate benchmark for $k = 8, 9, 10$, and hence, by adding independent coordinate outputs, for every $k \geq 8$. The same construction misses the benchmark at $k = 7$ by only $1/64$ in the high-noise coefficient. A broad collection of searches below eight bits also failed to produce a violation. This leads us to conjecture that the coordinate projection is optimal for $2 \leq k \leq 7$.

Throughout this section, entropies and mutual informations are measured in bits. Let

$$\begin{aligned} X &\sim \text{Unif}(\mathbb{F}_2^n), & Y &= X \oplus Z, \\ Z_i &\stackrel{\text{iid}}{\sim} \text{Bernoulli}(p), & 0 \leq p &\leq \frac{1}{2}. \end{aligned}$$

with Z independent of X , and write

$$h_2(p) = -p \log_2 p - (1-p) \log_2 (1-p), \quad C(p) = 1 - h_2(p).$$

A. Problem Formulation and Known Results

For $n \geq k$, define

$$M_k(n, p) := \max_{f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k} I(f(X); Y). \quad (23)$$

The coordinate projection $D_k(x) = (x_1, \dots, x_k)$ attains $kC(p)$. The vector extension asks whether

$$M_k(n, p) \leq kC(p) \quad \text{for every } n \geq k. \quad (24)$$

Besides $k = 1$, coordinate optimality is known for $k = n - 1$, including binary-input memoryless output-symmetric channels [28]. Chandar and Tchamkerten characterized the positive-rate asymptotics and gave counterexamples from the $[15, 11, 3]$ Hamming and $[23, 12, 7]$ Golay codes, asking for constructions with $k < 11$ [27].

Proposition 11 (Affine maps obey the coordinate bound). *If $f(x) = Ax + b$ has rank $r \leq k$, then*

$$I(f(X); Y) \leq rC(p) \leq kC(p).$$

Proof. Uniformity makes Z independent of Y and gives $I(f(X); Y) = r - H(AZ)$. For r independent columns of A , indexed by J ,

$$H(AZ) \geq H(AZ \mid Z_{J^c}) = rh_2(p). \quad \square$$

B. Coset Quantizers and the High-Noise Expansion

Let $H \in \mathbb{F}_2^{(n-k) \times n}$ have full row rank and $C = \ker H$ have dimension k . Choose a minimum-weight leader e_a for each syndrome, fixing ties, and define the coset quantizer

$$Q(x) = x \oplus e_{HX} \in C. \quad (25)$$

Identify C with $\mathbb{F}_2^k$. For uniform X , $E = e_{HX}$ is uniform over the leaders and independent of $Q(X)$; hence

$$I(Q(X); Y) = n - H(E \oplus Z). \quad (26)$$

For any fixed quantizer $Q: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$, write $U = Q(X)$ and set

$$B(Q) = \sum_{j=1}^n \mathbb{E} \left[\left(\mathbb{E}[(-1)^{X_j} \mid U] \right)^2 \right]. \quad (27)$$

With $\rho = 1 - 2p$, as $\rho \rightarrow 0$ the high-noise expansion is

$$I(U; Y) - kC(p) = \frac{\rho^2}{2 \ln 2} (B(Q) - k) + O(\rho^4). \quad (28)$$

To see this, for each output of positive probability put $m_j(u) = \mathbb{E}[(-1)^{X_j} \mid U = u]$. The density of $Y \mid U = u$ relative to the uniform measure is

$$1 + \rho \sum_j m_j(u) (-1)^{y_j} + O(\rho^2).$$

Taylor expansion of relative entropy and Walsh orthogonality give the coefficient $B(Q)/(2 \ln 2)$. The expansion is analytic and even in ρ : changing its sign complements every output coordinate without changing mutual information. Subtracting the expansion of $kC(p)$ proves (28). For coset quantizers, translation within each fiber gives

$$B(Q) = B(E) := \sum_{j=1}^n \left(\mathbb{E}[(-1)^{E_j}] \right)^2.$$

Thus $B(E) > k$ gives a counterexample for all $p < 1/2$ sufficiently close to $1/2$.

C. Shortened-Hamming Counterexamples

Identify nonzero vectors in $\mathbb{F}_2^4$ with $1, \dots, 15$ through their binary expansions. For $s = 1, \dots, 6$, let H_s have columns $s + 1, \dots, 15$, so

$$C_s = \ker H_s = [15 - s, 11 - s, 3], \quad n = 15 - s, \quad k = 11 - s.$$

Indeed, columns 8, 9, 10, 12 give rank four, while the columns are distinct and nonzero and $7 \oplus 8 \oplus 15 = 0$. Writing e_a for the

unit vector indexed by column a , choose the minimum-weight leaders

$$e_a = \begin{cases} 0, & a = 0, \\ \mathbf{e}_a, & a \in \{s+1, \dots, 15\}, \\ \mathbf{e}_8 \oplus \mathbf{e}_{8 \oplus a}, & a \in \{1, \dots, s\}. \end{cases} \quad (29)$$

Both columns in the last line are retained, so these are valid minimum-weight leaders. Among the 16 leaders, coordinate 8 occurs $s+1$ times, its s partners twice each, and each other coordinate once. Thus

$$B_s := B(E) = \frac{(7-s)^2 + 36s + 49(14-2s)}{64},$$

and consequently

$$B_s - k = \frac{s^2 - 12s + 31}{64}, \quad (30)$$

which is positive for $s = 1, 2, 3$.

Proposition 12 (Failure for every $k \geq 8$). *For every integer $k \geq 8$, there is a quantizer $f : \mathbb{F}_2^{k+4} \rightarrow \mathbb{F}_2^k$ such that*

$$I(f(X); Y) > kC(p)$$

for all $p < 1/2$ sufficiently close to $1/2$.

Proof. For $k = 10, 9, 8$, respectively, (30) gives $B_s - k = 5/16, 11/64, 1/16 > 0$; apply (28). For $k > 8$, append $k - 8$ independent coordinates to the eight-bit quantizer. Their information adds $(k-8)C(p)$, preserving the strict excess. $\square$

Table I summarizes numerical evaluations of $\Delta_k(p) = I(Q(X); Y) - kC(p)$. The displayed interval endpoints and maxima are numerical estimates; the preceding proof does not rely on them.

D. Searches Below Eight Bits

For $k = 7, 6, 5$, the same construction gives $B_s - k = -1/64, -1/16, -5/64$, respectively. At $k = 7$, exhaustive enumeration of all 1365 Hamming shortenings and 451,920 minimum-weight leader tables gave the maximum $B = 447/64 = 6.984375$. Numerical checks of the evaluated candidates across noise levels found no violation.

Additional searches covered 1941 puncturings of the $[15, 7, 5]$ BCH code; all 651 seven-dimensional codes between $\text{RM}(1, 4)$ and $\text{RM}(2, 4)$; 203 sampled Golay shortenings; 345,000 distinct-column parity-check searches with redundancies $5, \dots, 8$; and all 255 seven-dimensional quotients of the eight-bit Hamming quantizer, up to output relabeling. We also tested repeated columns and nonminimum coset representatives, with tie optimization where applicable. No violation was found at the tested crossover probabilities.

For 128 uniformly sampled codebooks of 2^7 words in $\mathbb{F}_2^{15}$, using fixed random nearest-neighbor tie assignments, the mean $B(Q)$ was 6.379431, with range $[6.342219, 6.423068]$. No sampled quantizer violated the benchmark at the tested probabilities. Heuristic tie optimization on 64 additional codebooks raised the mean coefficient to 6.6954, with maximum 6.71365, still below the shortened-Hamming value. These searches provide evidence, rather than an optimality proof.

E. A Seven-Bit Threshold Conjecture

Appending independent coordinate outputs propagates any counterexample to every larger output size. Together with the constructions and searches above, this motivates the following conjecture.

Conjecture 13 (Seven-bit threshold). *For every $2 \leq k \leq 7$, every $n \geq k$, every $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$, and every $0 \leq p \leq 1/2$,*

$$I(f(X); Y) \leq kC(p). \quad (31)$$

Equality is attained by a coordinate projection.

By the same padding argument, proving the conjecture for $k = 7$ would settle the entire range. A first target is $B(Q) \leq 7$ for every seven-bit quantizer, which controls the leading high-noise coefficient. The scalar theorem cannot simply be iterated: conditioning on earlier output bits generally destroys the uniform product input distribution required for the conditional information bounds.

VII. CONCLUSION

We proved that a coordinate is the most informative Boolean function of a uniform binary vector observed through a BSC. The proof combines monotone sorting, entropy flow, an entropy-constrained edge envelope, and spectral information carried by pivotal sets. Its key global feature is the growth implication: a positive advantage over a dictator would have to increase with channel correlation and therefore cannot be reconciled with the noiseless endpoint.

The vector-valued problem exhibits a different behavior. Shortened Hamming quantizers violate the coordinate benchmark for every fixed $k \geq 8$, whereas the structured and random searches, concentrated on $k = 7$, found no counterexample below eight bits. This leaves a concrete threshold problem. A particularly useful first step would be to prove the sharp high-noise bound $B(Q) \leq 7$ for every seven-bit quantizer and characterize its equality cases. Possible approaches to a complete result include a conditional form of the scalar theorem or a multiway entropy-flow inequality, since the ordinary chain rule does not preserve the uniform cube measure after conditioning.

GENERATIVE AI DISCLOSURE

The proof of the single-bit result, including its detailed mathematical derivations, was generated by generative AI systems; the authors' role was high-level direction, feedback, and final review. The multibit extension was entirely human-directed: the authors formulated the problem, selected the code-based constructions and computational questions, and interpreted the evidence and conjectures. Generative AI systems assisted with developing and checking proofs, implementing and verifying computations, and refining the exposition. The authors reviewed the complete manuscript and accept full responsibility for its mathematical claims, citations, and final text.

TABLE I
SHORTENED-HAMMING QUANTIZERS: EXACT HIGH-NOISE EXCESSES AND NUMERICALLY ESTIMATED POSITIVE INTERVALS AND MAXIMA.

k	code	$B_s - k$	numerical p -interval	$\max_p \Delta_k(p)/k$
10	[14, 10, 3]	5/16	(0.088658, 1/2)	4.48440×10^{-3}
9	[13, 9, 3]	11/64	(0.152276, 1/2)	1.83320×10^{-3}
8	[12, 8, 3]	1/16	(0.265563, 1/2)	3.21308×10^{-4}

APPENDIX A SCALAR EDGE AND ENTROPY INEQUALITIES

This appendix proves the scalar comparisons behind Lemma 2 and Proposition 9. Centering and convexity give the common entropy-budget bounds, and profile estimates control their angular correction. Entropy contraction and a mean correction give the influential-coordinate criterion.

A. Convex Lower Bound for Centered Edges

For edge values $p, q \in (-1, 1)$, define

$$\begin{aligned} d_{\text{edge}} &= \frac{|p - q|}{2}, & H_e &= \frac{h(p) + h(q)}{2}, \\ a(p, q) &= \frac{(p - q)(g(p) - g(q))}{4}, \\ j(p, q) &= \frac{(\arcsin p - \arcsin q)^2}{4}. \end{aligned}$$

Write $a = a(p, q)$, $j = j(p, q)$, and, for $k > 0$, define

$$\theta_{\text{edge}}(k) = \arcsin(\tanh k), \quad \text{corr}(F(k)) = k - \frac{\theta_{\text{edge}}(k)^2}{\tanh k}.$$

We prove

$$\begin{aligned} a(p, q) &\geq d_{\text{edge}} F^{-1}(H_e/d_{\text{edge}}), \\ a(p, q) - j(p, q) &\geq d_{\text{edge}} \text{corr}(H_e/d_{\text{edge}}). \end{aligned} \quad (\text{S2})$$

For $d_{\text{edge}}, H_e > 0$, both right-hand sides are convex in (d_{edge}, H_e) and decrease with H_e . At $d_{\text{edge}} = 0$, with $H_e > 0$, they extend continuously by zero.

1) *Centering property*: Assume $p > q$, and write $p = \tanh(v + k)$, $q = \tanh(v - k)$, where $k > 0$. Then $a = d_{\text{edge}} k$. With $M = \cosh(2v)$ and $C_k = \cosh(2k)$,

$$\begin{aligned} d_{\text{edge}} &= \frac{\sinh(2k)}{M + C_k}, \\ H_e &= \frac{1}{2} \log[2(M + C_k)] - \frac{v \sinh(2v) + k \sinh(2k)}{M + C_k}. \end{aligned}$$

For $v > 0$, the derivative of H_e/d_{edge} has the sign of $\log[2(\cosh x + C_k)] - x \coth x$, where $x = 2v$. It is nonnegative:

$$\begin{aligned} \log[2(\cosh x + C_k)] &\geq x + 2 \log(1 + e^{-x}) \\ &\geq x + \frac{2}{e^x + 1} \\ &\geq x + \frac{2x}{e^{2x} - 1} = x \coth x. \end{aligned}$$

Here we used $\log(1 + y) \geq y/(1 + y)$ and $e^x - 1 \geq x$. Evenness therefore gives

$$H_e/d_{\text{edge}} \geq F(k). \quad (\text{S3})$$

For $b_{\text{edge}} = \sinh k$ and $x = b_{\text{edge}}/\cosh v$, the half-angle identity gives

$$\arcsin p - \arcsin q = 2 \arctan x,$$

$$\frac{j}{d_{\text{edge}}} = \frac{b_{\text{edge}}}{\sqrt{1 + b_{\text{edge}}^2}} (1 + x^2) \left(\frac{\arctan x}{x} \right)^2.$$

The logarithmic derivative of $(1 + x^2)(\arctan x/x)^2$ is $2(x - \arctan x)/[x(1 + x^2) \arctan x] > 0$. Thus $x \leq b_{\text{edge}}$ gives

$$j/d_{\text{edge}} \leq \theta_{\text{edge}}(k)^2/\tanh k. \quad (\text{S4})$$

2) *Averaging by convexity*: Put $L_k = \log(2 \cosh k)$. Then

$$F'(k) = -\frac{L_k}{\sinh^2 k}, \quad (F^{-1})'(F(k)) = -\frac{\sinh^2 k}{L_k}.$$

The positive ratio increases with k , since $2L_k > \tanh^2 k$. Hence F^{-1} is decreasing and convex.

Similarly,

$$\begin{aligned} \frac{d}{dk} \left(k - \frac{\theta_{\text{edge}}(k)^2}{\tanh k} \right) &= \left(1 - \frac{\theta_{\text{edge}}(k)}{\sinh k} \right)^2, \\ \text{corr}'(F(k)) &= -\frac{(\sinh k - \theta_{\text{edge}}(k))^2}{L_k}. \end{aligned}$$

The latter positive ratio increases: its derivative has the sign of $2L_k \sinh k - (\sinh k - \theta_{\text{edge}}(k))^2 > 0$, since $2L_k > 1$. Thus corr is decreasing and convex, and is nonnegative by its zero limit as $k \downarrow 0$.

Combining these facts with (S3)–(S4) proves (S2). For $\psi \in \{F^{-1}, \text{corr}\}$, the perspectives $d_{\text{edge}} \psi(H_e/d_{\text{edge}})$ are convex and decrease with H_e , so Jensen applies to (S2).

Let $u = T_\rho f$, with f nonconstant monotone Boolean and $0 < \rho < 1$. Each active coordinate has $d_{\text{edge}} > 0$, $\mathbb{E} d_{\text{edge}} = \rho a_i$, and $\mathbb{E} H_e = \mathbb{E} h(u)$. For any upper bound $H_{\text{budget}} \geq \mathbb{E} h(u) > 0$, Jensen and decrease in entropy give

$$\begin{aligned} \mathbb{E} a &\geq \rho a_i v_i, \\ \mathbb{E}(a - j) &\geq \rho a_i [v_i - c(v_i)], \\ F(v_i) &= H_{\text{budget}}/(\rho a_i). \end{aligned}$$

Summing proves Lemma 2; inactive coordinates contribute zero.

B. Monotonicity of the Entropy Profile

Lemma 14 (Profile monotonicity). *The function $vF(v)$ strictly decreases for $v > 0$.*

The function $(1 - r^2)g(r) - rh(r)$ vanishes at both endpoints of $[0, 1]$ and has second derivative $-r/(1 - r^2) < 0$. Hence

$$h(r) < \frac{(1 - r^2)g(r)}{r}, \quad 0 < r < 1.$$

With $r = \tanh v$, we have $r - v(1 - r^2) > 0$, since $\sinh(2v) > 2v$. Substitution gives

$$\begin{aligned} \frac{d}{dv}[vF(v)] &= \frac{h(r)[r - v(1 - r^2)]}{r^2} - \frac{v^2(1 - r^2)}{r} \\ &< \frac{v(1 - r^2)}{r^3}(r - v) < 0, \end{aligned}$$

since $r < v$.

Choose an upper height $0 < v_{\text{cut}} \leq \ell_\rho$. For $0 < v \leq v_{\text{cut}}$ and $a = F(\ell_\rho)/F(v)$, monotonicity gives $a/v \leq F(\ell_\rho)/[v_{\text{cut}}F(v_{\text{cut}})]$. Also $c(v) \leq v$, since $\text{corr} \geq 0$. The LSI branch of Lemma 4 now gives

$$\text{yield}(v) \leq \rho + pK_{\text{LS}}\rho^2 \frac{F(\ell_\rho)}{v_{\text{cut}}F(v_{\text{cut}})} \quad (0 < v \leq v_{\text{cut}}). \quad (\text{S5})$$

C. Relative Rates of the Entropy Profile and Angle Correction

Lemma 15 (The profile pays for the angle correction). *Define the logarithmic decay rate $Q(v) = -F'(v)/F(v)$. Then*

$$\begin{aligned} Q(v) - \frac{1}{v} &> \frac{3}{2}[1 - c'(v)] \quad (v > 0), \\ Q(v) - \frac{1}{v} &> 2[1 - c'(v)] \quad (0 < v \leq 2). \end{aligned} \quad (32)$$

Proof. Write $r_v = \tanh v$, $\tau_v = \text{sech } v$, $\theta_v = \arcsin r_v$, and $s(v) = \theta_v / \sinh v$. Then $0 < \theta_v < v < \sinh v$, and

$$s' = \frac{\tau_v^2 - s}{r_v} < 0, \quad c' = 2s - s^2, \quad c'' = 2(1 - s)s' < 0.$$

Here $0 < s < 1$ and $\theta_v > r_v\tau_v$. Thus c increases, $1 - c' = (1 - s)^2$, and $c(v)/v$ decreases by concavity and $c(0) = 0$.

We also use

$$\frac{3z}{2 + \sqrt{1 - z^2}} \leq \arcsin z \leq \frac{\pi z}{2 + \sqrt{1 - z^2}} \quad (0 \leq z \leq 1). \quad (33)$$

Set $z = \sin t$. The quotient $t(2 + \cos t)/\sin t$ increases from 3 to π on $(0, \pi/2)$: the numerator of its derivative vanishes at zero and has derivative $2 \sin t(t - \sin t) > 0$.

With $q_v = e^{-2v}$, the entropy formula and $\log(1 + q_v) \leq q_v$ give

$$Q(v) = \frac{4q_v[v + \log(1 + q_v)]}{(1 - q_v)[(1 + q_v)\log(1 + q_v) + 2vq_v]} \geq \frac{4v}{2v + 1}.$$

Also $Q(v) > \coth v$: for

$$\begin{aligned} \mathcal{H}(v) &= \cosh v h(\tanh v), \\ D(v) &= v \coth v - \log(2 \cosh v), \end{aligned}$$

we have $D' = (\tanh v - v)/\sinh^2 v < 0$, $\lim_{v \rightarrow \infty} D(v) = 0$, and $\mathcal{H}' = -\sinh v D < 0$. Now use $F = \mathcal{H}/\sinh v$.

The power series show that $(\cosh v - 1)/v^2$ and $\sinh v/v$ increase, and geometric tail bounds give

$$\begin{aligned} \cosh 1 - 1 &< \frac{1/2}{1 - 1/12} < \frac{3}{5}, \\ \sinh 1 - 1 &< \frac{1/6}{1 - 1/20} < \frac{1}{5}, \\ \cosh 2 - 1 &< \frac{2}{1 - 1/3} = 3. \end{aligned}$$

Also $\cosh 4 < 33$, using $e^4 < 55$ from Section F-G. For $0 < v \leq 1$, $\cosh v \leq 1 + 3v^2/5$ and (33) yield

$$1 - s \leq \frac{2v^2}{5 + 2v^2}, \quad Q(v) - 1/v > \coth v - 1/v > \frac{5v}{18} > \frac{v}{4}.$$

The second inequality uses $v \cosh v - \sinh v = \int_0^v t \sinh t dt \geq v^3/3$ and $\sinh v < 6v/5$. Finally $(5 + 2v^2)^2 - 32v^3 \geq 25 - 12v^2 > 0$, so $v/4 > 2[2v^2/(5 + 2v^2)]^2$.

For $1 \leq v \leq 2$, use $\cosh v \leq 1 + 3v^2/4$, giving $1 - s \leq v^2/(2 + v^2)$. Multiplying

$$\frac{4v}{2v + 1} - \frac{1}{v} - 2 \frac{v^4}{(2 + v^2)^2},$$

by the positive factor $v(2v + 1)(2 + v^2)^2$ gives

$$v^3(15v - 4v^2 - 8) + 4(3v + 1)(v - 1) > 0.$$

Here $15v - 4v^2 - 8 \geq 3$ on $[1, 2]$. This proves the stronger bound.

For $2 \leq v \leq 4$, the bound $\cosh v \leq 1 + 2v^2$ gives $1 - s \leq 4v^2/(3 + 4v^2)$. Multiplying

$$\frac{4v}{2v + 1} - \frac{1}{v} - \frac{3}{2} \frac{16v^4}{(3 + 4v^2)^2},$$

by the positive factor $v(2v + 1)(3 + 4v^2)^2$ gives

$$16v^4(v - 2)(v - 3/2) + 16v^3(2v - 3) + 3(4v^2 - 6v - 3) > 0.$$

Each term is nonnegative, and the last quadratic is increasing from one. For $v \geq 4$, simply use $Q(v) - 1/v \geq 2 - 2/(2v + 1) - 1/v > 3/2$ and $(1 - s)^2 < 1$. $\square$

D. Propagation of the Local Cutoff

For $0 \leq r, a \leq 1$, write

$$M_{\text{loc}}(r, a) = (1 - r^2)h(ar) - (1 - ar^2)h(r).$$

Yu's threshold-propagation lemma [11, Lemma 4.9], in sign means, states that $M_{\text{loc}}(r_1, a) \geq 0$, with $0 < r_1 < 1$, implies $M_{\text{loc}}(r, a) \geq 0$ for $0 \leq r \leq r_1$. Since $M_{\text{loc}}(r, \cdot)$ is concave and $M_{\text{loc}}(r, 1) = 0$, a check at (r_1, a_0) , with $a_0 \in [0, 1]$, therefore covers all $0 \leq r \leq r_1$ and $a_0 \leq a \leq 1$.

E. The Influential-Coordinate Criterion

For Boolean f and $0 < \rho < 1$, fix a coordinate and write $a = |\widehat{f}(\{i\})|$, $m = \mathbb{E}f$, $\delta = 1 - \rho^2$. Then $|m| + a \leq 1$.

For a cube field v taking values in $[-1, 1]$, define $\text{Ent}_\phi(v) = \mathbb{E}\phi(v) - \phi(\mathbb{E}v)$. The standard entropy contraction following from cube LSI [30], [31] is

$$\text{Ent}_\phi(T_\rho v) \leq \rho^2 \text{Ent}_\phi(v). \quad (\text{S6a})$$

Here $\text{Ent}(w) = \mathbb{E}[w \log w] - (\mathbb{E}w) \log(\mathbb{E}w)$, with $0 \log 0 = 0$. Apply $\text{Ent}(T_\rho w) \leq \rho^2 \text{Ent}(w)$ to $w = 1 \pm v$ and use $2 \text{Ent}_\phi(v) = \text{Ent}(1 + v) + \text{Ent}(1 - v)$.

Noise the chosen bit first, then apply (S6a) to the remaining coordinates. The intermediate values are $\pm \rho$ on pivotal edges and ± 1 otherwise; pivotal probability is at least a , and the section means are $m \pm \rho a$. Thus

$$\mathbb{E}h(T_\rho f) \geq \rho^2 a h(\rho) + \frac{\delta}{2} [h(m + \rho a) + h(m - \rho a)]. \quad (\text{S6})$$

The required mean correction is

$$\phi(m) + \frac{\delta}{2}[h(m + \rho a) + h(m - \rho a)] \geq \delta h(\rho a). \quad (\text{S7})$$

The difference vanishes with its first derivative at $m = 0$. Its second derivative has the sign of

$$[1 - (m + a)^2][1 - (m - a)^2] + (1 - \rho^2)a^2(1 - m^2 - a^2) \geq 0$$

on $|m| + a < 1$. Convexity proves (S7) there, and continuity covers $|m| + a = 1$.

Combining (S6)–(S7) proves CK whenever

$$(1 - \rho^2)h(\rho a) - (1 - \rho^2 a)h(\rho) \geq 0. \quad (\text{S8})$$

Section A-D extends a successful cutoff to larger a and smaller correlations.

APPENDIX B

AN ENTROPY TANGENT FOR THE ANGULAR ERROR

The angular comparison is the common starting point for both growth arguments in Section IV. We prove its pointwise form (8) by viewing the angular error as part of an entropy variable and taking a supporting line at the dictator value.

To prove (8), fix $5/4 \leq \lambda \leq \pi/2$ and set

$$A_{\text{sc}}(x) = x \arcsin x, \quad \Psi(x) = \phi(x) + (\arcsin x - \lambda x)^2/2.$$

For $0 < t < \pi/2$, put $x = \sin t$, $c = \cos t$, and $g = \text{atanh } x$. Differentiation with respect to t gives

$$K_\lambda := \ddot{A}_{\text{sc}}\dot{\Psi} - \dot{A}_{\text{sc}}\ddot{\Psi} = g(1 + c^2) - x(t^2 + 2) + \lambda^2(x - c^3 t).$$

Since $x - c^3 t > 0$, it suffices to prove $K_{5/4} > 0$. The elementary bounds

$$\begin{aligned} \arctan y &\leq \frac{y(15 + 4y^2)}{15 + 9y^2}, \\ \text{atanh } y &\geq \frac{y(15 - 4y^2)}{15 - 9y^2} \quad (0 \leq y < 1) \end{aligned}$$

follow by differentiation and equality at zero. Substituting $y = \tan(t/2)$ gives

$$\frac{t}{x} \leq \frac{19 + 11c}{3(1 + c)(4 + c)}, \quad \frac{g}{x} \geq \frac{11 + 19c}{3(1 + c)(1 + 4c)}.$$

These imply

$$\frac{16K_{5/4}}{x} \geq \frac{(1 - c)Q(c)}{9(1 + c)(4 + c)^2(1 + 4c)},$$

where

$$\begin{aligned} Q(c) &= 375(2 - 6c - 7c^2)^2 + 164(1 - c)^5 \\ &\quad + 740c(1 - c)^4 + c^2(1 - c)^3 \\ &\quad + 25c^3(1 - c)^2 + 5925c^4(1 - c) + 8625c^5. \end{aligned}$$

Thus $K_\lambda > 0$ for $0 < t < \pi/2$. Moreover,

$$\left(\frac{\dot{\Psi}}{\dot{A}_{\text{sc}}} \right)' = -\frac{K_\lambda}{\dot{A}_{\text{sc}}^2} < 0.$$

The ratio tends to $\pi/2 - \lambda \geq 0$ as $t \uparrow \pi/2$, so $\dot{\Psi} > 0$ and $d^2 A_{\text{sc}}/d\Psi^2 > 0$.

Now let $r = \tanh \ell$, $\ell \geq 31/20$, and $\lambda = \arcsin(r)/r$. Since $e^{31/10} > 22$, we have $r^2 > 5/6$, and the alternating sine bound gives

$$\frac{\sin(5r/4)}{r} \leq \frac{5}{4} - \frac{125}{384}r^2 + \frac{3125}{122880}r^4 < 1.$$

Thus $5/4 < \lambda \leq \pi/2$, the upper bound following from the endpoint chord of $\arcsin$. At $x = r$, $\Psi(r) = \phi(r)$ and

$$\frac{dA_{\text{sc}}}{d\Psi}(r) = \frac{\arcsin r + r/\sqrt{1 - r^2}}{\text{atanh } r} = \alpha.$$

The supporting line is therefore $A_{\text{sc}}(x) - A_{\text{sc}}(r) \geq \alpha[\phi(x) - \phi(r) + (\arcsin x - \lambda x)^2/2]$, which is (8). Evenness and continuity cover negative x and the endpoints.

APPENDIX C

CHANNEL-PARAMETER BOUNDS IN THE INTERMEDIATE REGIME

The intermediate-correlation proof reduces to three inequalities that depend only on the channel. We establish the profile comparison needed for the log-sum step, the angular-energy reserve, and the bound that absorbs the output bias. These are precisely the inputs used to close Proposition 3.

Lemma 16 (The profile, energy, and mean reserves). *Let $31/20 \leq \ell \leq 7/2$, with the channel parameters and profiles defined in Sections III and IV. Then*

$$\log \frac{vF(v)}{\ell F(\ell)} \geq \beta[H(\ell) - H(v)] \quad (0 < v \leq \ell), \quad (34)$$

$$\theta^2 - rH(\ell) - \frac{\eta}{2} \geq 0, \quad (35)$$

$$\gamma - \frac{2\theta^2}{\sinh^2 \ell} - \frac{\lambda}{\alpha} \geq r\ell. \quad (36)$$

Proof. By Lemma 15, c increases, while $c(v)/v$ and $s(v) = \arcsin(\tanh v)/\sinh v$ decrease. Also λ increases, and $\alpha > 2$ increases because it averages the increasing function $\text{sech } t + \cosh t$ on $[0, v]$. Hence γ increases. We use (33) and the elementary bounds

$$z - \frac{z^3}{3} \leq \arctan z \leq z - \frac{z^3}{3(1 + z^2)} \quad (z \geq 0). \quad (37)$$

Fixed constants below follow from Section F-G and finite exponential sums.

a) Profile bound.: The identity

$$\beta(\ell) = \frac{2}{c(\ell)} \left(1 + \frac{1}{\gamma(\ell)} \right)$$

shows that β decreases. At $\ell = 31/20$, Appendix B gives $\lambda > 5/4$. Together with $r > 19/21$ and $\alpha > 2$, this yields $c(31/20) > 475/336 > 7/5$, $\gamma(31/20) > 5/2$, and $\beta(31/20) < 2$.

At $\ell = 2$, $155/21 < e^2 < 37/5$ gives $r < 27/28$, $\sinh 2 > 18/5$, and, by (37), $\theta = \pi/2 - 2 \arctan(e^{-2}) > 13/10$. Thus $c(2) > 7/4$, $\lambda(2) > 4/3$, $\alpha(2) > 49/20$, and $\gamma(2) > 13/4$. Consequently $\beta(2) < 136/91 < 3/2$. Lemma 15 now gives $Q(v) - 1/v \geq \beta(\ell)[1 - c'(v)]$: use its coefficient two when $\ell \leq 2$, and three halves when $\ell \geq 2$. Integration proves (34).

b) *Energy bound.*: Using $rH(\ell) = r\ell - \theta^2$, (35) becomes

$$\frac{c(\ell)}{\ell} \left[\frac{3}{2} + \frac{1}{2(\gamma+1)} \right] \geq 1.$$

Both factors decrease. At $\ell = 7/2$, $32 < e^{7/2} < 35$ gives

$$\begin{aligned} \theta &> \frac{\pi}{2} - \frac{1}{16}, & \lambda &< \frac{8}{5}, \\ \alpha &< \frac{8/5 + 35/2}{7/2} = \frac{191}{35}, & \gamma &< 9. \end{aligned}$$

Since $c > \theta^2$, the product exceeds

$$\frac{31}{20} \frac{(\pi/2 - 1/16)^2}{7/2} > 1,$$

using $\pi > 157/50$.

c) *Mean bound.*: We prove (36) for $\ell \geq \ell_0 = 3/2$. Put $\tau = \text{sech } \ell$ and $s = s(\ell)$. The exponential bounds and (37) give

$$\begin{aligned} \frac{19}{21} &< \tanh \ell_0 < \frac{77}{85}, & \frac{75}{32} &< \cosh \ell_0 < \frac{5}{2}, \\ \frac{19}{9} &< \sinh \ell_0 < \frac{77}{36}, & \frac{9}{8} &< \theta(\ell_0) < \frac{8}{7}. \end{aligned} \quad (38)$$

Hence $81/154 < s(\ell_0) < 72/133$, and the decrease of s gives $2s^2 < 3/5$ throughout this range. Define

$$Y(\ell) = c(\ell) + \theta \cosh \ell - r\ell^2 - \frac{6}{5}\ell.$$

Then $Y/\ell = \gamma - r\ell - 6/5$, and, with $T = \theta \cosh \ell - r$, differentiation gives

$$Y'' = T \left[1 - \frac{2(1-s)}{\sinh^2 \ell} \right] + 2\tau^2 \ell(r\ell - 2).$$

Here $r > 9/10$, $\cosh \ell > 7/3$, $\sinh^2 \ell > 40/9$, and $\lambda > 6/5$ by (33). Thus $T > 81/50$, and its bracket exceeds $11/20$. Also $\ell\tau^2$ decreases from a value below $3/10$, while $r\ell - 2 > -13/20$. Hence

$$Y'' > \frac{81}{50} \frac{11}{20} - \frac{39}{100} > \frac{1}{2}.$$

The same endpoint bounds give $Y(\ell_0) > 3/16$ and $Y'(\ell_0) > -1/5$. Taylor's inequality, with $t = \ell - \ell_0 \geq 0$, yields

$$Y(\ell) > \frac{3}{16} - \frac{t}{5} + \frac{t^2}{4} = \frac{1}{4} \left(t - \frac{2}{5} \right)^2 + \frac{59}{400} > 0.$$

Thus $\gamma - r\ell > 6/5$.

Finally, (33) gives

$$\frac{\lambda}{\alpha} = \frac{\ell\theta}{r(\theta + \sinh \ell)} \leq \frac{\pi\ell}{(\pi+1)r + 2\sinh \ell}.$$

For $Z(\ell) = (\pi+1)r + 2\sinh \ell - (5\pi/3)\ell$, (38) and $\pi < 22/7$ give

$$\begin{aligned} Z'' &= 2\sinh \ell - 2(\pi+1)r\tau^2 > 4 - \frac{522}{343} > 0, \\ Z'(\ell_0) &> \frac{16}{25} + \frac{14}{3} - \frac{110}{21} > 0, \\ Z(\ell_0) &> \frac{19}{21} + \frac{38}{9} - \frac{67}{42} \frac{22}{7} > 0. \end{aligned}$$

Thus $Z > 0$ and $\lambda/\alpha < 3/5$. Consequently

$$\gamma - r\ell - 2s^2 - \frac{\lambda}{\alpha} > \frac{6}{5} - \frac{3}{5} - \frac{3}{5} = 0.$$

This proves (36). $\square$

APPENDIX D

THE SMALL-CORRELATION REGIME

We prove CK for $0 \leq r = \rho \leq r_1 = \rho_* := \tanh(31/20)$ using the local/Fourier-gap approach [6], [10], [11]. Proposition 9 handles large singleton coefficients; the remaining functions are controlled by a singleton-energy bound and quadratic entropy minorants. Entropy contraction then extends the bound to the lowest correlations.

Write $W_1(v) = \sum_i \hat{v}(\{i\})^2$, $a = \max_i |\hat{f}(\{i\})|$, and set

$$a_0 = r_0 = \frac{5}{8}, \quad M = 1 - a_0 + a_0^2 = \left(\frac{7}{8} \right)^2.$$

The case $r = 0$ is immediate.

A. Singleton-Energy Bound

For Boolean f , put

$$a_i = \hat{f}(\{i\}), \quad a = \max_i |a_i|, \quad W = \sum_i a_i^2 = W_1(f).$$

We prove

$$a \leq \frac{5}{8} \implies W \leq 1 - a_0 + a_0^2 = M. \quad (39)$$

Orient a coordinate with coefficient $a \geq 0$ and let $s = (f_+ + f_-)/2$, $d = (f_+ - f_-)/2$. Since $s^2 + d^2 = 1$, $d \in \{-1, 0, 1\}$, and $\mathbb{E}d = a$, Parseval gives

$$W = a^2 + W_1(s) \leq a^2 + \mathbb{E}s^2 \leq a^2 + 1 - a. \quad (40)$$

For $3/8 \leq a \leq 5/8$, the last expression is at most its common endpoint value M .

1) *A Sign-Sum Bound.* For independent uniform signs X_i , let $Z = \sum_i b_i X_i$. We show

$$\sum_i b_i^2 = 1, \quad \max_i b_i^2 \leq \frac{1}{5} \implies \mathbb{E}|Z| < \frac{7}{8}. \quad (41)$$

The Hermite interpolant to $\sqrt{y}$ at $9/16, 4, 9$, matching values and first derivatives, is

$$\begin{aligned} D_{\text{mom}} p(y) &= 637696y^5 - 17928800y^4 \\ &\quad + 191446315y^3 - 1008169920y^2 \\ &\quad + 3660186285y + 1260006516, \end{aligned} \quad (42)$$

where $D_{\text{mom}} = 4042912500$. The interpolation remainder gives, for $y > 0$ away from the contacts and some $\xi > 0$,

$$\sqrt{y} - p(y) = \frac{(\sqrt{\cdot})^{(6)}(\xi)}{6!} (y - 9/16)^2 (y - 4)^2 (y - 9)^2 \leq 0,$$

since $(\sqrt{\cdot})^{(6)}(\xi) = -945/(64\xi^{11/2}) < 0$. Continuity includes the contacts and zero, so $|Z| \leq p(Z^2)$.

Write $\sigma_j = \sum_i b_i^j$ and $t = \sigma_4$. The coefficient bound and Jensen's inequality with weights b_i^2 give

$$\begin{aligned} 0 &\leq t \leq \frac{1}{5}, & \sigma_6 &\leq \frac{t}{5}, \\ \sigma_{10} &\leq \frac{\sigma_8}{5}, & \sigma_8 &\geq t^3. \end{aligned}$$

Expanding the moments in $\mathbb{E}e^{xZ} = \prod_i \cosh(b_i x)$ yields

$$\begin{aligned} D_{\text{mom}} \mathbb{E}p(Z^2) &= 3487476486 - 214438410t \\ &\quad + 1507452800t^2 \\ &\quad + (1459014320 - 4285317120t)\sigma_6 \\ &\quad - 2928765440\sigma_8 + 5060755456\sigma_{10}. \end{aligned}$$

The σ_6 coefficient is positive on $[0, 1/5]$. After substituting $\sigma_{10} \leq \sigma_8/5$, the σ_8 coefficient is $-9583071744/5$. Thus $\sigma_6 \leq t/5$ and $\sigma_8 \geq t^3$ imply $D_{\text{mom}} \mathbb{E}p(Z^2) \leq T_{\text{mom}}(t)$, where

$$\begin{aligned} T_{\text{mom}}(t) &= 3487476486 + 77364454t \\ &\quad + 650389376t^2 - \frac{9583071744}{5}t^3. \end{aligned}$$

The derivative T'_{mom} is concave and positive at both endpoints:

$$\begin{aligned} T'_{\text{mom}}(0) &= 77364454 > 0, \\ T'_{\text{mom}}(1/5) &= \frac{13440810318}{125} > 0. \end{aligned}$$

Hence $T_{\text{mom}}(t) \leq T_{\text{mom}}(1/5)$, and

$$7D_{\text{mom}} - 8T_{\text{mom}}(1/5) = \frac{119582002252}{625} > 0.$$

This proves (41).

If $a \leq 3/8$ and $W \geq M$, take $b_i = a_i/\sqrt{W}$. Then $\sum_i b_i^2 = 1$, $\max_i b_i^2 \leq 9/49 < 1/5$, and

$$\sqrt{W} = \mathbb{E} \left[f \sum_i b_i X_i \right] \leq \mathbb{E} \left| \sum_i b_i X_i \right| < \frac{7}{8}.$$

This contradicts $W \geq M$. Together with (40), this proves (39).

B. The Local Cutoff

For the local margin M_{loc} of Section A-D, Section D-G gives

$$M_{\text{loc}}(r_1, 5/8) = (1 - r_1^2)h(5r_1/8) - (1 - 5r_1^2/8)h(r_1) > 0.$$

Concavity in a , equality at $a = 1$, and the propagation result in Section A-D extend this to $a \geq 5/8$, $r \leq r_1$. Proposition 9 then applies.

C. Mean-Sensitive Energy Bound

The preceding estimate gives

$$a \leq a_0 \implies W_1(f) \leq M. \quad (43)$$

Section D-B covers $a \geq a_0$. For $a \leq a_0$, form the balanced lift

$$G(1, x) = f(x), \quad G(-1, x) = -f(-x).$$

Its singleton coefficients are $m = \mathbb{E}f$ and those of f , so $|m| \leq a_0$ implies

$$m^2 + W_1(f) = W_1(G) \leq M. \quad (44)$$

For $|m| \geq a_0$, entropy contraction (S6a) gives, when $r > 0$,

$$I_f(r) \leq r^2 h(m) < r^2/2 \leq \phi(r),$$

since $h(m) \leq L - a_0^2/2 - a_0^4/12 < 7/10 - a_0^2/2 - a_0^4/12 < 1/2$. Hence assume $|m| \leq a_0$.

D. Tangent Quadratic Minorants

For $0 < t < 1$, define

$$\begin{aligned} A_t &= \frac{\log(2/(1+t))}{(1-t)^2}, \\ z_t &= 2t - \frac{g(t)}{A_t}, \\ Q_t(s) &= A_t(1-s)(1+s-z_t). \end{aligned}$$

Then $Q_t(t) = h(t)$, $Q'_t(t) = h'(t)$, and $Q_t(1) = h(1) = 0$. Moreover,

$$Q_t(s) \leq h(s) \quad (0 \leq s \leq 1). \quad (45)$$

Indeed, $D_t = h - Q_t$ satisfies $D_t(t) = D'_t(t) = D_t(1) = 0$, and $D''_t(s) = 2A_t - 1/(1-s^2)$ strictly decreases. Necessarily $D''_t(t) > 0$; otherwise $D_t(1) < 0$. Thus D_t decreases to zero on $[0, t]$, then rises and falls to zero on $[t, 1]$.

Take $t_0 = 2/5$, $t_1 = 2/3$, $A_j = A_{t_j}$, $z_j = z_{t_j}$, and

$$\begin{aligned} \beta &= \frac{1/A_0 - 1/A_1}{z_1 - z_0}, \\ \alpha &= \frac{1}{A_0} + \beta z_0, \\ c_{\text{ent}}(r) &= \frac{1}{\alpha - \beta r^2}. \end{aligned}$$

Section D-G verifies $z_0 < r_0^2 \leq r_1^2 < z_1$. For $r \in [r_0, r_1]$, write $r^2 = \mu z_0 + (1-\mu)z_1$, with $0 \leq \mu \leq 1$. Combining (45) with weights $\mu c_{\text{ent}}/A_0$ and $(1-\mu)c_{\text{ent}}/A_1$, which sum to one, yields

$$h(s) \geq c_{\text{ent}}(r)(1-s)(1+s-r^2). \quad (46)$$

E. From Entropy to Fourier Energy

Writing $W_k = \sum_{|S|=k} \widehat{f}(S)^2$, we have

$$\mathbb{E}|T_r f| \geq \mathbb{E}[f T_r f] = \sum_{k=0}^n r^k W_k.$$

Apply (46) to $s = |T_r f|$. Since h is even, Parseval and $r^{k+2} - r^{2k} \geq 0$ for $k \geq 2$ give

$$\mathbb{E}h(T_r f) \geq c_{\text{ent}}(r)[1 - r^2 - (1 - r^2)m^2 - r^2(1 - r)W_1].$$

Insert (44) and $\phi(m) \geq m^2/2$:

$$\begin{aligned} \mathbb{E}h(T_r f) + \phi(m) &\geq c_{\text{ent}}(r)(1-r)(1+r - Mr^2) \\ &\quad + [1/2 - c_{\text{ent}}(r)(1 - 2r^2 + r^3)]m^2. \end{aligned} \quad (47)$$

The mean coefficient has the sign of $\alpha - 2 + (4 - \beta)r^2 - 2r^3$, which increases on $[r_0, 1]$ because $\beta < 1$. At r_0 , the bounds $\alpha > 53/40$, $\beta < 17/20$ give a value exceeding $43/640$, so this term is nonnegative.

F. Positivity of the Remaining Gap

It remains to prove $B_0(r) \geq 0$ on $[r_0, r_1]$, where

$$B_0(r) = (1-r)(1+r-Mr^2) - (\alpha - \beta r^2)h(r).$$

Differentiation gives

$$B_0''''(r) = \frac{2[\alpha(1+3r^2) - \beta(r^4 - 3r^2 + 6)]}{(1-r^2)^3}. \quad (48)$$

Its numerator increases with r^2 . Together with $B_0'''(r_0) < 0$, this implies that B_0''' changes sign at most once, from negative to positive; hence B_0'' has no interior maximum. The bounds in Section D-G therefore give $B_0'' < 1/3$ on $[r_0, 4/5]$ and $B_0'' < 0$ on $[4/5, r_1]$. Since $B_0'(r_0) < -1/16$,

$$B_0'(r) < -\frac{1}{16} + \frac{4/5 - 5/8}{3} = -\frac{1}{240} \quad (r_0 \leq r \leq 4/5),$$

and B_0' decreases thereafter. Thus $B_0(r) \geq B_0(r_1) > 0$, proving CK on $[r_0, r_1]$ by (47).

For the remaining case $a, |m| \leq a_0$, the fixed checks also give the starting slack

$$c_{\text{ent}}(r_0)(1-r_0)(1+r_0-Mr_0^2) + r_0^2/2 - L > 0.$$

Thus $I_f(r_0) < r_0^2/2$. Applying entropy contraction to $T_r f = T_{r/r_0}(T_{r_0} f)$ covers the rest:

$$I_f(r) \leq (r/r_0)^2 I_f(r_0) \leq r^2/2 \leq \phi(r) \quad (0 \leq r \leq r_0).$$

G. Fixed Comparisons

For $1 \leq y \leq 2$, put $z = (y-1)/(y+1)$ and

$$\ell_-(y) = 2 \sum_{j=0}^{11} \frac{z^{2j+1}}{2j+1},$$

$$\ell_+(y) = \ell_-(y) + \frac{2z^{25}}{25(1-z^2)}.$$

The logarithm series gives

$$\ell_-(y) \leq \log y \leq \ell_+(y).$$

For other positive rational arguments, extract a power of two. Applied to the defining logarithms of A_j, z_j, α, β , these give

$$\alpha > \frac{53}{40}, \quad 0 < \beta < \frac{17}{20}, \quad z_0 < r_0^2, \quad r_1^2 < z_1.$$

For the endpoint checks, use

$$r_1 = \frac{e^{31/10} - 1}{e^{31/10} + 1}, \quad g(r_1) = \frac{31}{20},$$

$$h(r_1) = \log(1 + e^{-31/10}) + \frac{31/10}{1 + e^{31/10}}.$$

Bound the exponential by Section F-G. Substituting these rational enclosures and $\ell_{\pm}$ into B_0 , its first three derivatives, and M_{loc} , with outward bounds throughout, certifies the following margins.

Fixed expression	Bound
$M_{\text{loc}}(r_1, 5/8)$	$> 1/1400$
$B_0'(5/8)$	$< -17/250$
$B_0''(5/8)$	$< 7/30$
$B_0'''(5/8)$	< -1
$B_0''(4/5)$	$< -1/9$
$B_0''(r_1)$	$< -1/10$
$B_0(r_1)$	$> 1/4000$
Starting contraction slack at $r_0 = 5/8$	$> 1/500$

APPENDIX E

ENDPOINT REDUCTION FOR THE HEIGHT PARAMETER

The tail argument requires a scalar comparison at every active coordinate height. The logarithmic-Sobolev estimate covers heights below $\ell/2$; above that point, the Parseval yield has no interior maximum. We reduce the remaining comparison to the join and the local cutoff, which are verified in Appendix F.

A. Shape Properties and Endpoint Reduction

Recall $c(v) = \arcsin^2(\tanh v)/\tanh v$ and $F(v) = h(\tanh v)/\tanh v$. Set

$$J = 1/F, \quad \Lambda(v) = \log(2 \cosh v),$$

$$\mathcal{H}(v) = \cosh v h(\tanh v).$$

The proof of Lemma 15 gives $c'' < 0$, $0 < c' < 1$, and $\mathcal{H}' < 0$ on $v > 0$. Since $J' = \Lambda/\mathcal{H}^2$, also $J'' > 0$.

For $\ell = \ell_\rho \geq 7/2$, $r = \rho$, and $p = (\lambda + 1/\alpha)^2$, the Parseval yield is

$$Y_P(v) = \frac{rc(v) + p[d_2/2 + (d_1 - d_2/2)F(\ell)/F(v)]}{v}.$$

Section E-C proves that it has no interior maximum. The LSI comparison covers $v \leq z = \ell/2$. Set

$$a_{\text{cut}} = \frac{1}{1 + (8/3)e^{-\ell}}, \quad F(V) = \frac{F(\ell)}{a_{\text{cut}}}.$$

The decrease of F gives a unique $0 < V < \ell$. Section F-B shows that a counterexample must have $a_i < a_{\text{cut}}$, and hence $\ell_i < V$. Thus only $Y_P(z), Y_P(V) < B/\ell$ remain when $V > z$; if $V \leq z$, the LSI comparison suffices.

B. Entropy-Profile Decay

The cutoff proof uses

$$Q_F(v) := -F'(v)/F(v) > 8/5 \quad (v > 0). \quad (49)$$

Substituting $q = e^{-2v}$ and clearing positive denominators reduces this to

$$vq(1+4q) + (5q-2+2q^2)\log(1+q) > 0.$$

Only a negative logarithmic coefficient needs consideration. Using $\log(1+q) < q$, it suffices that $v-2+(4v+5)e^{-2v} > 0$. This is immediate for $v \geq 2$. On $[0, 2]$, $(2-v)e^{2v}/(4v+5)$ is maximized at $(3+\sqrt{65})/8 \in (11/8, 7/5)$, giving the upper bound $5e^{14/5}/84 < 1$ (Section F-G).

For this cutoff, integration gives

$$\frac{8}{5}(\ell - V) < \int_V^\ell Q_F(v) dv = \log(1 + (8/3)e^{-\ell}) < \frac{8}{3}e^{-\ell}.$$

Hence $0 < \ell - V < (5/3)e^{-\ell}$.

C. Endpoint Principle for the Original Yield

We prove the curvature bound

$$\frac{-c''(v)[vJ'(v) - J(v)]}{J''(v)} - [c(v) - vc'(v)] < \frac{5}{32}, \quad (50)$$

$$v > 0.$$

It implies that $Y_P(v) = [rc(v) + A_0 + A_1J(v)]/v$, with $A_0, A_1 > 0$, has no interior maximum if $A_0/r \geq 5/32$: its derivative has the sign of

$$A_1 - \frac{A_0 + r[c(v) - vc'(v)]}{vJ'(v) - J(v)},$$

and the fraction strictly decreases by (50). For the Parseval yield, $A_0 = pd_2/2$ and $A_1 = p(d_1 - d_2/2)F(\ell)$; Appendix F-C verifies $A_0/r > 3/16 > 5/32$ for $\ell \geq 7/2$.

a) *Entropy curvature.*: Write $r_v = \tanh v$, $\delta_v = 1 - r_v^2$, and $D = vJ' - J > 0$. Chebyshev's integral inequality gives

$$\frac{h(r_v)}{\delta_v} = \int_0^1 \frac{ds}{(1+s)(1-r_v^2s^2)} \leq L \int_0^1 \frac{ds}{1-r_v^2s^2} = \frac{Lv}{r_v}.$$

For $0 < v \leq 1$, this is at most $L \coth 1 < 1$, since $e^2 > 7$ and $L < 3/4$. Thus $\mathcal{H}'' = (h(r_v) - \delta_v)/\sqrt{\delta_v} < 0$. Consequently $\mathcal{H}^{-2}$ is increasing and convex, and $J''' = (\Lambda \mathcal{H}^{-2})'' > 0$. Hence

$$D = \int_0^v tJ''(t) dt < \frac{v^2}{2} J''(v) \quad (0 < v \leq 1). \quad (51)$$

We extend this to

$$\frac{D}{J''} < \frac{v}{2} \quad (v > 0). \quad (52)$$

Only $v \geq 1$ remains. Put $q = e^{-2v}$, $T = \log(1+q)/q$, and $K = 2v + (1+q)T$. Then

$$J = \frac{1-q}{qK}, \quad K' = 2T,$$

$$qD = 1 - \frac{1-q+2T}{K} + \frac{(1-q^2)T^2}{K^2},$$

$$0 < T' = 2[T - 1/(1+q)] < q.$$

The last bound is the trapezoid estimate for $1/(1+q)$. Using $q < 1/7$, $1-q/2 \leq T < 1$, $K > 2$, and

$$qK < e^{-2v}(2v + 3/2) \leq (7/2)e^{-2} < 1/2,$$

and dropping the positive derivative of $(1-q^2)T^2$, we obtain

$$(qD)' \geq \frac{1}{K^2} \left[2T(1-q+2T) - K(2q+2T') \right. \\ \left. - \frac{4(1-q^2)T^3}{K} \right] > 0 :$$

the three terms are respectively greater than 4, less than 2, and less than 2. Since $(qD)' = q(vJ'' - 2D)$, this proves (52).

b) *Angular curvature.*: Put $w = \arcsin(\tanh v)/\sinh v$, $\tau_v = \operatorname{sech} v$, and $k = -c'' > 0$. The angle bound (33) gives $w \geq 3\tau_v/(2+\tau_v) > \tau_v^2$. Using $w' = (\tau_v^2 - w)/r_v < 0$, differentiation yields

$$vw'' - w' \geq \frac{\tau_v(1-\tau_v)}{r_v^2(2+\tau_v)} [(3+\tau_v)r_v + v\{4 - (1+\tau_v)^3\}] > 0.$$

Indeed, $r_v \geq v\tau_v$ bounds the bracket below by $v(3 - 2\tau_v^2 - \tau_v^3) > 0$. Thus $f = 1 - w$ is increasing and concave in v^2 , with $f(0) = 0$. For $0 < t \leq v$, $f(t) \geq (t/v)^2 f(v)$ and $f'(t) \geq (t/v)f'(v)$. Since $k = 2ff'$,

$$c(v) - vc'(v) = \int_0^v tk(t) dt \geq \frac{v^2}{5} k(v). \quad (53)$$

Maximizing the quadratic in $k = 2(1-w)(w - \delta_v)/r_v$ gives $k \leq r_v^3/2 < 1/2$. Therefore

$$\frac{kD}{J''} - (c - vc') < k(v) \left(\frac{v}{2} - \frac{v^2}{5} \right) \leq \frac{5}{32},$$

which proves (50).

APPENDIX F CANONICAL HIGH-CORRELATION COMPARISONS

This appendix establishes the high-correlation case of Lemma 10 by verifying the baseline conditions (17) and the yield comparison (19) for $\ell \geq 7/2$. After excluding large influences by the local criterion, we check small heights and the two endpoints of the remaining height interval. Appendix E extends those endpoint checks to the whole interval.

A. Reduction to Three Comparisons

Write $r = \rho = \tanh \ell$, $x = e^{-\ell}$, $L = \log 2$, $\theta = \arcsin r$, and $P = \pi^2/4$. Recall

$$\lambda = \theta/r, \quad \alpha = (\theta + \sinh \ell)/\ell, \quad (54)$$

$$\gamma = \lambda\alpha, \quad d = \ell/2 + 5/4.$$

As before, $\omega(s) = \gamma s/(\gamma + s)$, $d_1 = [\omega(d) - \omega(1)]r^2$, $d_2 = [\omega(d) - \omega(2)]r^4$, and $d_b = d_1 - d_2/2$. The retained-variance baseline is

$$p = (\lambda + 1/\alpha)^2, \quad b = \lambda^2(2 + 1/\gamma),$$

$$C_d = p\omega(d) - b, \quad B = 2\theta^2 + C_d\phi(r)/L.$$

The local criterion covers $a \geq a_{\text{cut}} = 1/[1 + (8/3)x]$. A possible violation therefore has all active heights $v < V$, where $F(V) = F(\ell)/a_{\text{cut}}$. Below we verify the baseline conditions, the LSI comparison through $z = \ell/2$, and the Parseval comparisons at z and V . The endpoint principle of Appendix E fills $[z, V]$ when $V > z$; otherwise the LSI comparison suffices.

B. Local Cutoff

For $\ell \geq 7/2$, put $x = e^{-\ell} < 1/32$ and

$$a_{\text{cut}} = \frac{1}{1 + (8/3)x}. \quad (55)$$

Write $h_b(t) = -t \log t - (1-t) \log(1-t)$ and set

$$\begin{aligned} \kappa &= \frac{4}{3+8x}, & \eta &= \frac{3x}{4(1+x^2)}, \\ t &= \frac{x^2}{1+x^2}, & q &= \kappa x(1+\eta), \\ A &= \frac{1}{2x^2} + \frac{3}{4x} + 1 + \frac{x^2}{2}. \end{aligned}$$

Thus $t = (1-r)/2$, $q = (1-ra_{\text{cut}})/2$. The margin $(1-r^2)h(ra_{\text{cut}}) - (1-a_{\text{cut}}r^2)h(r)$, divided by the positive factor $4\kappa x^3/(1+x^2)^2$, is

$$\Lambda = \frac{h_b(q)}{\kappa x} - Ah_b(t).$$

The reciprocal midpoint and trapezoid bounds

$$\frac{2s}{2+s} \leq \log(1+s) \leq s - \frac{s^2}{2(1+s)} \quad (s \geq 0)$$

give $h_b(q) \geq q \log(1/q) + q - q^2/(2-q)$ and $(1+\eta) \log(1+\eta) \leq \eta + \eta^2/2$. Hence

$$\begin{aligned} \frac{h_b(q)}{\kappa x} &\geq (1+\eta)(\ell - \log \kappa) + 1 \\ &\quad - \frac{\eta^2}{2} - \frac{\kappa x(1+\eta)^2}{2-q}. \end{aligned}$$

Since $\eta \leq 3x/4$, the last two errors satisfy

$$\begin{aligned} \frac{\eta^2}{2} + \frac{\kappa x(1+\eta)^2}{2-q} &\leq \frac{9}{32}x^2 + \frac{x(4+3x)^2}{12(2+4x-x^2)} \\ &\leq \frac{2}{3}x + \frac{1}{5}x^2. \end{aligned}$$

Indeed, the rational term is at most $2x/3 - x^2/12$, because

$$(8-x)(2+4x-x^2) - (4+3x)^2 = x(6-21x+x^2) > 0$$

for $x < 1/4$, and $9/32 - 1/12 < 1/5$. Using $h_b(t) = \log(1+x^2) + 2\ell x^2/(1+x^2)$, the upper logarithm bound gives

$$Ah_b(t) \leq \frac{1}{2} + \frac{3}{4}x + \frac{3}{4}x^2 + \frac{2A\ell x^2}{1+x^2}.$$

The discarded remainder is $x^3[2x(1+x^2)-3]/[8(1+x^2)] < 0$. Since $1+\eta - 2Ax^2/(1+x^2) = -x[3/(4(1+x^2)) + x]$, subtraction gives

$$\begin{aligned} \Lambda &\geq \frac{1}{2} - (1+\eta) \log \kappa - \frac{17}{12}x - \frac{19}{20}x^2 \\ &\quad - \ell x \left[\frac{3}{4(1+x^2)} + x \right]. \end{aligned} \quad (56)$$

The same logarithm bound gives $\log(4/3) < 7/24$, while $\ell e^{-\ell}$ decreases. Thus

$$\begin{aligned} (1+\eta) \log \kappa &< \left(1 + \frac{3}{128}\right) \frac{7}{24} < \frac{1}{3}, \\ \frac{17}{12}x + \frac{19}{20}x^2 &< \frac{17}{384} + \frac{19}{20480} < \frac{1}{20}, \\ \ell x \left[\frac{3}{4(1+x^2)} + x \right] &< \frac{7}{64} \left(\frac{3}{4} + \frac{1}{32} \right) < \frac{3}{32}. \end{aligned}$$

Therefore $\Lambda > 1/2 - 1/3 - 1/20 - 3/32 = 11/480 > 0$. The local margin is concave in a and vanishes at $a = 1$, so (22) holds throughout $[a_{\text{cut}}, 1]$.

C. Canonical Penalty and Preliminary Bounds

The fixed bounds in Section F-G give $x < 1/33$, $r > 99/100$, $\theta > 3/2$, and $\lambda^2 < P < 5/2$, using $\arcsin r < \pi r/2$. Since $\theta + \sinh \ell > 1/(2x)$,

$$\gamma > \frac{3e^\ell}{4\ell} > \ell + \frac{7}{2} = 2d + 1. \quad (57)$$

The second inequality follows because $3e^\ell - 4\ell^2 - 14\ell$ and its first two derivatives are positive on this half-line.

The exact identities

$$\begin{aligned} p\omega(1) - b &= -\lambda^2, \\ p\omega(2) - b &= -\frac{\lambda^2}{\gamma + 2}, \end{aligned} \quad (58)$$

$$C_d = \lambda^2 \left[d - 2 - \frac{(d-1)^2}{\gamma + d} \right].$$

give $C_d > 0$, since $\gamma(d-2) > 1$, and $C_d < (5/2)(d-2)$, $b < (5/2)(2+1/7) < 6$. Since $0 < 1 - 1/(2L) < 1/3$, the mean condition follows from

$$\begin{aligned} \gamma - b - C_d \left(1 - \frac{1}{2L} \right) &> \ell + \frac{7}{2} - 6 - \frac{5}{6} \left(\frac{\ell}{2} - \frac{3}{4} \right) \\ &= \frac{1}{6} + \frac{7}{12} \left(\ell - \frac{7}{2} \right) > 0. \end{aligned}$$

The endpoint-principle coefficient satisfies

$$\begin{aligned} \frac{pd_2}{2r} &= \frac{p}{2}(d-2) \frac{\gamma}{\gamma+d} \frac{\gamma}{\gamma+2} r^3 \\ &> \frac{3}{16} > \frac{5}{32}, \end{aligned}$$

using $p/2 > 1$, $d-2 \geq 1$, both fractions $> 1/2$, and $r^3 > 3/4$. Also $0 < d_2 < d_1$, hence $d_b > 0$.

a) *Entropy estimate.*: Put $E = h(r) - L(1-r^2) \geq 0$, using $\phi(r) \leq Lr^2$. For every $\ell \geq 3/2$,

$$E \leq x^2(2\ell - 5/3). \quad (59)$$

Indeed, the entropy formula, $q = x^2$, and $\log(1+q)/q \leq (2+q)/[2(1+q)]$ give

$$2\ell - \frac{E}{q} - \frac{5}{3} \geq \frac{4(L-2/3) - 11q/6 + 5q^2/6}{(1+q)^2} > 0.$$

Its numerator exceeds $8/81 - 11/120 > 0$, since $L - 2/3 > 2/81$ and $q < 1/20$.

D. Small Heights: Paired Spectral Costs

Let $z = \ell/2$. The line-angle LSI comparison is

$$\begin{aligned} M_{\text{small}} &= \frac{B}{\ell} - r - \frac{pK_{\text{LS}}r^2F(\ell)}{zF(z)}, \\ K_{\text{LS}} &= \frac{\gamma^2 e^{2d-3}}{2(\gamma+d)(\gamma+1)}. \end{aligned}$$

Set $a_d = d - 1$, $k = e^{-1/2}$, $T = e^\ell F(\ell)/F(z)$, and $f_d = C_d/\lambda^2$. By (58) and $\phi(r)/L = r^2 - E/L$,

$$\begin{aligned} \frac{\ell M_{\text{small}}}{\theta^2} &= d - \frac{\ell}{c(\ell)} - kT \\ &\quad - \frac{a_d(a_d - kT)}{\gamma + d} - \frac{f_d E}{Lr^2}. \end{aligned} \quad (60)$$

The bounds $t/(1+t) \leq \log(1+t)$ and $(1+t)\log(1+t) \leq t + t^2/2$ give

$$\begin{aligned} \frac{2\ell + 1}{(\ell + 1 + x/2)(1+x)} &\leq T \leq \frac{2\ell + 1 + x^2/2}{(\ell + 1)(1+x)} \\ &\leq 2 - \frac{1}{\ell + 1}. \end{aligned}$$

The lower bound exceeds $5/3$, because

$$\ell - 2 - 5\ell x - \frac{15}{2}x - \frac{5}{2}x^2 \geq \frac{3}{2} - \frac{25}{32} - \frac{5}{2048} > 0.$$

Here the numerator increases with ℓ and decreases with x . Using $3/5 < k < 5/8$ and $c(\ell) > \theta^2 > 9/4$,

$$\begin{aligned} 1 < kT &< \frac{5}{8} \left(2 - \frac{1}{\ell + 1} \right), \\ d - \frac{\ell}{c(\ell)} - kT &> \frac{\ell}{18} + \frac{5}{8(\ell + 1)} \geq \frac{1}{3}, \end{aligned}$$

the last difference from $1/3$ being $(2\ell - 3)(2\ell - 7)/[72(\ell + 1)]$. Also $\gamma + d > 4(d - 1)(d - 2)$: with $s = \ell - 7/2 \geq 0$, the cubic Taylor lower bound gives

$$\begin{aligned} 3e^\ell - (4\ell^3 - 6\ell^2 - 8\ell) \\ > 29 + 2s + \frac{27}{2}s^2 + \frac{25}{2}s^3 > 0. \end{aligned}$$

Together with (57), this yields

$$\frac{a_d(a_d - kT)}{\gamma + d} < \frac{a_d(a_d - 1)}{\gamma + d} < \frac{1}{4}.$$

Finally, (59) gives

$$\frac{f_d E}{Lr^2} < 2(d - 2)(2\ell - 5/3)e^{-2\ell} < \frac{1}{96},$$

using $f_d < d - 2$, $Lr^2 > 1/2$, and monotonicity of the product, whose value at $7/2$ is below $2(16/3)/32^2 = 1/96$. Thus

$$M_{\text{small}} > \frac{\theta^2}{\ell} \left(\frac{1}{3} - \frac{1}{4} - \frac{1}{96} \right) = \frac{7\theta^2}{96\ell} > 0.$$

Since $vF(v)$ decreases and $c(v) \leq v$, this covers $0 < v \leq z$ and gives $B/\ell > r$.

E. Half-Height Join and Angular Saving

Write $a_z = F(\ell)/F(z)$. The original Parseval margin at the join has the exact form

$$\begin{aligned} M_{\text{join}} &:= Bz/\ell - rc(z) - p[d_2/2 + d_b a_z] \\ &= r[c(\ell) - c(z)] - \frac{\lambda^2 r^4}{2(\gamma + 2)} + \frac{C_d}{2} \left[1 - r^4 - \frac{h(r)}{L} \right] \\ &\quad - p d_b a_z. \end{aligned}$$

The multiplier identities also give

$$\begin{aligned} p d_b &= C_d(r^2 - r^4/2) + \lambda^2 \left[r^2 - \frac{r^4}{2(\gamma + 2)} \right] \\ &\leq C_d/2 + P \leq P d/2. \end{aligned} \quad (61)$$

To prove that $e^v[P - c(v)]$ increases, write $t = \arcsin(\tanh v)$. Then

$$\begin{aligned} (t + \cos t)^2 - c(v) - c'(v) \\ = (1 - \sin t)(t/\sin t - \cos t)^2 + \sin t \cos^2 t > 0, \end{aligned}$$

and $t + \cos t < \pi/2$, so $c + c' < P$. At $v_0 = \log(10/3)$, $4/\pi > 5/4$ and $\arctan(3/10) \geq 3/10 - (3/10)^3/3$ give

$$2t/\pi < 16/25, \quad c(v_0)/P < (16/25)^2(109/91) < 1/2.$$

Since $z \geq 7/4 > v_0$, $P - c(z) > (5P/3)\sqrt{x}$. Combining this with $c(\ell) \geq \theta^2 \geq P - 2\pi x$ gives

$$c(\ell) - c(z) > \frac{5P}{3}\sqrt{x} - 2\pi x.$$

Using $h(r) < (2\ell + 1)x^2$, $a_z = xT < 2x$, (61), and (57), and dropping $C_d(1 - r^4)/2 > 0$, gives

$$\begin{aligned} \frac{M_{\text{join}}}{P\sqrt{x}} &> \frac{5r}{3} - \left(\frac{7}{6}\ell + \frac{5}{4} + \frac{8}{\pi} \right) e^{-\ell/2} \\ &\quad - \frac{(d - 2)(2\ell + 1)}{2L} e^{-3\ell/2}. \end{aligned}$$

The first term increases and both losses decrease. At $7/2$, $r > 99/100$, $\sqrt{x} < 7/40$, $8/\pi < 8/3$, and $2L > 4/3$ bound the losses by $8(7/40)$ and $6(7/40)^3 < 1/25$. Hence

$$\frac{M_{\text{join}}}{P\sqrt{x}} > \frac{33}{20} - \frac{7}{5} - \frac{1}{25} = \frac{21}{100} > 0.$$

F. Cutoff Bound with Exact Low-Degree Multipliers

Let $F(V) = F(\ell)/a_{\text{cut}}$. By (49),

$$0 < \ell - V < \frac{5}{3}x.$$

The Parseval margin is

$$M_{\text{cut}} = \frac{B}{\ell}V - rc(V) - p[d_2/2 + d_b a_{\text{cut}}].$$

The identity

$$B - p d_1 - rc(\ell) = C_d[\phi(r)/L - r^2] = -C_d E/L$$

and $c' > 0$ give

$$M_{\text{cut}} \geq -C_d E/L + p d_b(1 - a_{\text{cut}}) - \frac{B}{\ell}(\ell - V).$$

Using $B \leq 2\lambda^2 r^2 + C_d$, the equality in (61), and (59),

$$\begin{aligned} \frac{M_{\text{cut}}}{x} &> C_d \left[\frac{8r^2(1 - r^2/2)}{3 + 8x} - \frac{(2\ell - 5/3)x}{L} - \frac{5}{3\ell} \right] \\ &\quad + \lambda^2 r^2 \left[\frac{8}{3 + 8x} \left(1 - \frac{r^2}{2(\gamma + 2)} \right) - \frac{10}{3\ell} \right]. \end{aligned}$$

Since $r^2 > 3/4$, $x < 1/32$, and $(2\ell - 5/3)e^{-\ell}$ decreases,

$$\begin{aligned} \frac{8r^2(1 - r^2/2)}{3 + 8x} &> \frac{32}{13} \frac{15}{32} > 1, \\ \frac{(2\ell - 5/3)x}{L} &< \frac{16}{3} \frac{1}{32} \frac{3}{2} = \frac{1}{4}, \\ \frac{5}{3\ell} &\leq \frac{10}{21} < \frac{1}{2}. \end{aligned}$$

Thus the first bracket exceeds $1/4$; the second exceeds $2(3/4) - 1 = 1/2$, using $\gamma > 7$ and $10/(3\ell) \leq 20/21 < 1$. Therefore

$$M_{\text{cut}} > x \left(\frac{C_d}{4} + \frac{\lambda^2 r^2}{2} \right) > 0.$$

Together with the local cutoff and the endpoint principle, this completes the height comparison for every $\ell \geq 7/2$.

G. Fixed Constants

For $0 < y < 14$, the fixed arithmetic uses

$$\begin{aligned} E_{12}(y) &:= \sum_{j=0}^{12} \frac{y^j}{j!}, \\ E_{12}(y) &< e^y < E_{12}(y) + \frac{y^{13}}{13!(1 - y/14)}, \end{aligned}$$

and, for $|y| < 1$,

$$\begin{aligned} \log \frac{1+y}{1-y} &= 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1} + R_N(y), \\ |R_N(y)| &\leq \frac{2|y|^{2N+1}}{(2N+1)(1-y^2)}. \end{aligned}$$

Use $y = (q-1)/(q+1)$ for $\log q$; the low-correlation arithmetic takes $N = 12$. Five terms at $y = 1/3$, and Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, with four/five and two/three alternating terms respectively, give

$$3.14159 < \pi < 3.1416, \quad .69314 < \log 2 < .69315.$$

In particular, $25/8 < \pi < 22/7$, $9/4 < P < 5/2$, and $2/3 < L < 3/4$. The positive exponential sums give $e^3 > 20$, $e^{7/2} > 33$, and $3/5 < e^{-1/2} < 5/8$. Square-root bounds follow by squaring positive rational endpoints.

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